

Relation as Category

An Epistemological Investigation

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1. The History of the Concept of Relation

The present treatise is, like an earlier treatise on “Totalitet som Kategori” (Videnskabsernes Selskabs Skrifter, 1917), an attempt at the further working out of a single point in my book “Den menneskelige Tanke” (1910), in which it was my endeavour to give a presentation of the principal forms under which human thinking works, and, in connexion with these, of the principal problems on which it works. The connexion between forms of thought and problems of thought shows itself plainly in this, that the forms of thought lie at the basis of the questions that are put — questions which, when they are put in a definite and sharp form, become problems.

After interest in epistemology had been awakened anew at the close of the last century, it was above all the first fundamental propositions of human knowledge that drew attention to themselves. Their possible grounding or derivation was discussed, and so was the possibility of carrying them completely through in every domain. But such fundamental propositions contain certain fundamental concepts, which are applied more or less involuntarily, partly by so-called sound common sense (*sens commun*), the pre-scientific thinking, partly by scientific thinking, as the latter displays itself in the history of the sciences. Both in the questions thought puts, and in the answers it acknowledges, certain forms display themselves, which it is the task of the

doctrine of the categories, of descriptive epistemology, to bring out and to order. The two principal sources for the doctrine of the categories are the psychological study of “sound common sense” and the historical study of the development of science proper through its various stages; and these sources stand in constant reciprocal action with one another, in that science develops successively out of the involuntary reflection of ordinary consciousness, especially when a doubt has arisen, and in that, on the other hand, the train of thought and the results of science slowly gain influence on the form and content of ordinary consciousness, so that sound common sense is not the same at all times.

While the earlier treatise dealt with the concept of totality and the tendencies of thought connected with it, the present treatise shall seek to throw light on the concept of relation and the principles of inquiry that hang together with it. The two concepts point reciprocally to one another. The concept of totality asserts itself along two different paths. All knowledge consists in a comprehending together, an involuntary striving to form wholes where such are not immediately given. But the immediately given itself, the objects before us, can appear (and on closer investigation perhaps always does appear) as wholes, and they can be understood only by analysis of the connexion between their elements and with other wholes and elements. Human knowledge forms wholes where such are not immediately given, and dissolves them where they immediately lie before us. In both cases the concept of the whole shows itself to hang together with the concept of relation. Wholes can be formed only by objects being brought into definite relations to one another, and given wholes are understood only by the demonstration of the inner and outer relations that are decisive for their subsistence. Relation therefore shows itself to be an essential form of thought in every domain. To think is to find definite relations, to set one object in relation to another; and the most general relations show themselves to be continuity and discontinuity, likeness and difference. These general or fundamental categories assume more special forms in the formal categories (identity, analogy, negation, rationality), in the real categories (causality, totality, development) and in the ideal categories (the concepts of value). The closer demonstration of this is given in “Den menneskelige Tanke”.

The concept of relation will now in what follows receive illumination through an investigation of the way in which it asserts itself in the setting up and the discussion of the principal philosophical problems. In my University Programme of 1902 and in “Den menneskelige Tanke” (p. 246–254) I have maintained that there are four such problems: the psychological, the epistemological, the ethical and the cosmological, and have pointed out the analogies between them. —

An overview of the history of the doctrine of the categories (see “Den menneskelige Tanke” p. 140–153) shows a decided tendency to place the concept of relation in the first rank. In Aristotle, the first who set up a doctrine of the categories, relation (*πρὸς τι*) is, to be sure, only one among ten categories; but closer consideration shows that most (and at bottom all) the other Aristotelian categories express particular species of relation. Aristotle himself names as examples of relation: equality and inequality, cause and effect, knowledge and the known. But in later Greek thinking relation came to play an altogether decisive part, and here the overview given in “Den menneskelige Tanke” needs an important supplementing. Characteristic of this later Greek thinking, before mysticism made its influence felt, was the critical-sceptical direction it struck out upon. Plato stands, in spite of his speculative striving, as a precursor of this direction, partly by the objections he allows to be voiced against the doctrine of Ideas in “Parmenides”, partly in the weight he lays, in later dialogues, on knowledge expressing itself in the judgment, that is, in the determination of the mutual relations of concepts (cf. my “Bemærkninger om den platoniske Dialog Parmenides” in *Videnskabernes Selskabs filosofiske Meddelelser*, 1920). But while Plato was, in spite of everything, convinced that the truth had been found and was expressed in a world of eternal Ideas, the later Academic school came to the result that it had not merely not been found, but could not be found; what one could attain by the rational use of experience was only probability. The most considerable man of this school is Carneades (2nd cent. B.C.), who has rightly been called the mightiest mind in Greece in these centuries. It became of decisive significance for the concept of relation that he raised the question wherein our criterion for truth and reality consists. As against the Stoics, who rested content with the view that there were presentations

in which the truth could be grasped quite immediately (φαντασίαι καταληπτικαί)¹, he maintains that, since several mutually conflicting presentations can appear with this stamp, one cannot rely on any single presentation whatever. The validity of a presentation can rest only on its agreement with other presentations. It must not merely be ἀπερίστατος, not contradicted by other presentations, but it must find confirmation through its connexion with other presentations, just as physicians in their diagnoses do not hold to any single mark, but rely on the connexion between all the marks (συνδρομή τῶν σημείων). Further, there must in all important cases be required an “analysing review” of all the relations that have to do with presentations (so that the presentation becomes a φαντασία διεξωδευμένη).²

It would be wrong to call Carneades a sceptic. He does not believe that any presentation whatever has a claim to validity apart from its relation to other presentations; but he thereby really only points to the path which so-called sound common sense involuntarily takes when a doubt arises, and he points to the method of empirical science. In his account of Carneades’ doctrine, Sextus compares his method with that of the empirical physicians, and the conjecture has also been expressed that Carneades stands in debt to this school of medicine.³ In the criterion of truth maintained by Carneades the concept of relation plainly displays itself, and moreover in a way already hinted at by Plato (Republic X, p. 602 C–603 A), as we shall have occasion to mention at a later place.

If now the criterion of truth is the contradiction-free relation between well-tested and thoroughly thought-out presentations, then a task is thereby set that can never be absolutely solved. There are no limits to the number of relations in which a presentation must be investigated in order that it may stand its test. It shows itself here, what will show itself again and again in what follows, that the concept of relation conditions at one and the same time a limit and a task. By reason of this concept’s predominant significance in our knowledge, every understanding that is won expresses certain definite relations and holds only for them, is to that extent “relative”. But relation as category leads, on the other hand, always beyond every

¹On the interpretation of this expression see PAUL BARTH: *Die Stoa*, p. 66 f.

²*Sextus Empiricus* ed. Bekker, p. 225–232.

³VICTOR BROCHARD: *Les Sceptiques Grecs*. 1889, p. 183.

understanding won, in that it points to the importance or the necessity of seeking new relations which can make the determination of the object understood more complete, and thereby the understanding itself more deep-going. What offers resistance and arrests or limits thought, precisely thereby sets it new tasks, yields new impulses. Charles Renouvier has therefore rightly spoken of relation as method (*la méthode des relations*). In the history of the concept of relation, right down to the most recent time, the two tendencies contend — limit and task, resistance and work, conclusion and extension. And both tendencies spring from relation as category.

Already in antiquity this contention shows itself. The Academics after Carneades' time were mostly taken up with the conclusion, and thereby approached the Stoics and their belief in a truth given once for all. The Sceptics, on the other hand, held that when one can never have done with winning the full truth, one has no right to assert that it is. Besides, it may of course be asked by what criterion we can decide that the criterion maintained by Carneades is the right one. — While the Dogmatists reject the concept of relation and cling to certain presentations as definitive, the Sceptics maintain, on the basis of the concept of relation, that there is no truth, for a relative truth is no truth.

The most eminent of these Sceptics, Aenesidemus (probably in the first cent. B.C.), has given a systematic account of the various relations which in his opinion make true knowledge impossible. He points to the differences of living beings, particularly of men, as regards the faculty of apprehension, — to the difference between our various senses among themselves, whereby “one and the same thing” comes to stand as different for different senses, — to the way in which our states and our standpoints in space and time, particularly the distances from the object, condition our apprehension, — and to the influence of custom and law. In all he sets up 10 species of relation which render our knowledge uncertain. Sextus Empiricus (ed. Bekker, p. 10) shows that these 10 species (τρόποι) can be led back to three, in that either the apprehending and judging subject (ὁ κρίνων), or the apprehended and judged object (τὸ κρινόμενον), or else both at once, can determine the apprehension. And these three species of relation he leads back again to the general concept of relation (ὁ πρὸς τι τρόπος). Here for the first time relation is set up as a general category.

It has been wrongly supposed that in the part which the discussion of the criterion of reality plays towards the close of antiquity there is a sign of a weakening of the impulse to inquiry.⁴ It is precisely new and significant thoughts that come forward in the Academic and the Sceptical school, points of view which Plato and Aristotle in their intellectual enthusiasm had not allowed to come into their own. The insight that the decision as to what can be acknowledged as reality is determined partly by the relation of objects among themselves, partly by the relation between the objects and the apprehending and thinking subject, has passed over from the Academic and Sceptical school to the philosophy and natural science of the more recent age⁵, as will show itself in what follows. In antiquity itself the discussion of this question formed the introduction to a purely empirical science, particularly in the schools of medicine; and a scientific movement in an empirical direction was perhaps, as already remarked, contributory to the very origin of the question in principle. —

In more recent times it is above all criticism and positivism that have laid weight on the concept of relation. Kant, whose doctrine of the categories is the most significant since Aristotle, came to his philosophy through the insight that the category of relation is the most important of all, in that our knowledge everywhere aims at connecting objects through the determination of their mutual relations. It is two species of relation that become especially significant, relations of dependence and relations of likeness, and it is in the work of thinking such relations that knowledge reveals itself as synthesis.⁶ In Kant too the contention between the two points of view regarding the concept of relation asserts itself. He has a certain inclination to regard it as an imperfection in our knowledge that we place everything in relations, and he therefore understands a “thing in itself” as lying outside all relations. Some Kantians, especially William Hamilton, have stressed this still more strongly and have set it up as a thoroughgoing limitation of our knowledge that work must always be done by setting one object in relation to another. In a similar way Herbart declared that our relation-setting can give only an accidental apprehension of things, and that true reality does not stand in relations at all. But in Kant the essential point of view was nonetheless

⁴DEUSSEN: *Die Philosophie der Griechen*. 1911, p. 410.

⁵*Den menneskelige Tanke*, p. 104 f.

⁶*Om Kontinuiteten i Kants filosofiske Udviklingsgang* (Vid. Selsk. Skr. 1893) p. 53.

that what is characteristic of all science is the relation between objects expressed in definite laws. And from this point of view the necessity of relation for knowledge becomes precisely a source of fullness and richness. — The more relations that can be found and determined, the greater the unity and the greater the manifoldness there will be at one and the same time in our knowledge.

For Kant's speculative successors, relation became more and more a barrier which one strove to get beyond by seeking a highest, all-embracing unity, within which all relations, particularly all oppositions, were to stand as one-sided apprehensions. Fichte stands here as a characteristic transitional type. He has a clear eye for the fact that when we seek to find and to determine relations, it is a spiritual activity that displays itself. Without spiritual activity, no relation-setting: "Die reine Tätigkeit ist Bedingung des Beziehens," (*Wissenschaftslehre*², p. 244). Relation is nothing received or imposed, but presupposes an act of thought. And when by such an act a barrier is set, this means only that new activity is necessary. Spiritual work, the theoretical as well as the practical, seeks to reach beyond barriers, and complete freedom from barriers stands as the exalted ideal that can never be realized in any definite state. In Fichte's later philosophy, however, his problem became another. Instead of asking how we, through the setting of relations and through struggle with the barriers thus set, can attain an ever richer truth, he now asks, in Neoplatonic fashion, how there can be any striving after truth at all, when at the basis of everything there lies an absolute being.

Hegel's philosophy proceeded, in its historical orientation, from a standpoint on which human thoughts rested securely side by side, without any need asserting itself to determine their mutual relations. The arising of such a need is in his view a phenomenon of dissolution. It is an unhappy consciousness in which the mutual harmony of thoughts is broken. Then there is need of a strict spiritual work, in order that the individual thoughts, in spite of, or precisely by, their mutual oppositions, may come to stand as members of a great whole. This happens by means of the dialectical method, which at bottom means a systematization of concepts of relation. The individual thought leads beyond itself to other thoughts through the recognition of its limitation. The contradictions that arise through the isolation of thoughts hereby fall away. —

A critical investigation now shows, however, that it is at bottom Hegel himself who isolates the concepts, in order afterwards so victoriously to demonstrate their higher unity. Thus at the very beginning of his *Logic*⁷ he sets “being” as something on its own and “nothing” as something on its own, in order then to show that they cannot do without one another. Every being has, after all, its limit, beyond which there is nothing, if no new form of being can be found. And the concept of nothing means nothing other than precisely such a limit of being. It is the experience of limited being that Hegel resolves into two concepts standing in contradiction to one another. — But although Hegel himself thus often creates the contradictions his dialectic is to overcome, and although it also goes far too easily for him to pass from relation to relation, his philosophy is nonetheless a great expression of the truth that one can think existence only by working one’s way forward from relation to relation, and an expression of the fact that in this very work the highest that can be thought is present as a driving force.

With full consciousness the French Neo-Kantian Charles Renouvier has placed the concept of relation at the head of his doctrine of the categories. That everything is relative — he says — this strong word of scepticism was the last word of critical philosophy in antiquity, and it must be the first word of the modern method. And he finds a whole method contained in this category (*la méthode des relations*): the point is to determine our objects through as many relations as possible. — And criticism here meets with positivism. For Comte the concept of relation lies contained in the very concept of positive science, which everywhere seeks to put definite relations, expressed in laws, in place of the absolute but indefinite essences or forces of theology and metaphysics. From this side Comte expressly regards Kant as his precursor, just as he also meets with him in this, that the work of knowledge is conceived as consisting in connecting objects (*tout se réduit toujours à lier*) — partly through classification (*par similitude*), partly through derivation (*par filiation*).⁸ As Dauriac⁹ has shown, Comte’s classification of the sciences is laid out on the same principle as that which

⁷*Wissenschaft der Logik*. 1812. I, p. 22–26. — More recent Hegelians take a critical attitude here towards the master. See B. CROCE: *Cio che è vivo e ciò che è morto della filosofia di Hegel* (1907), p. 21–23. MC. TAGGART: *A Commentary on Hegels Logic* (1910), p. 17.

⁸*Discours sur l’esprit positif* (1844), p. 20.

⁹*Contingence et catégorie* (*Revue de Métaphysique et de Morale*. 1916), p. 499.

later lies at the basis of Renouvier's ordering of the categories, namely in such a way that one proceeds from the simple and fundamental to the complex and derived.

Goblot¹⁰ objects to every attempt to set up a doctrine of the categories that our knowledge is still too provisional, both as regards the final conclusions and as regards the fundamental principles, for a system of fundamental concepts to be capable of being set up. This objection does not touch Renouvier, who is clear that no rational proof can be given for the validity of the categories or the completeness of the list of categories. They are found by analysis of experience, are set up tentatively, and must always be confirmed anew by experience.¹¹ Renouvier therefore reproaches Kant with having attempted a deductive grounding of the doctrine of the categories. In peculiar opposition to this, Renouvier's disciple Hamelin¹², all the while holding fast to his predecessor's table of categories, has sought to give a dialectical derivation by showing that every category leads over into its opposite. But he confuses, as Hegel so often does, the correlation of concepts with a relation of mutual contradiction or opposition between them; such a relation comes forward only when the individual concepts are taken out of their correlation through an isolating abstraction, which is then cancelled again and thereby produces a semblance of dialectic. Hamelin's dialectical transitions are accordingly very artificial. — Beyond an analytical method one does not get, as Renouvier has very rightly seen, in the doctrine of the categories.

The categories are sought and found by analysis of the actual work our thought performs when definite questions are put to it. Every table of categories can therefore have only a provisional character, and the more special categories cannot be derived from the more general and comprehensive, although on closer investigation they may show themselves to be more special forms of them. As everywhere where we have to do with concepts drawn from experience (in this case from the actual work of thought), there is an inverse relation between content and extension. The more elements a concept contains, the narrower the circle of objects for which it holds. In the doctrine of the categories this acquires a particularly important significance through the relation between the concepts of value and the more universal (fundamental,

¹⁰*Traité de Logique* (1918), p. 138.

¹¹See Séailles: *La philosophie de Renouvier*, p. 91 f.

¹²*Essai sur les éléments principaux de la représentation*. (1907).

formal and real) categories. But, as I have sought to show in the fifth chapter of the treatise “Totalitet som Kategori”, since all value stands in relation to a certain whole and to that whole’s conditions of existence, there is nonetheless an inner connexion between the concepts of value and the other categories.

It is now the task of the present treatise to throw light on the concept of relation more closely by seeing it in its significance for the various philosophical problems. It will hereby be expedient not merely to hold to the concept of relation itself, but also to investigate the principle that corresponds to the concept, the principle of relation, which I (“Den menneskelige Tanke”, p. 255) have formulated thus: Every object shall be considered in as many relations as possible. Where we can find no relations, we obtain no knowledge. The relationless is therefore rightly declared by Kant to lie outside all science and to be only a purely negative concept. Our thinking works like a pair of compasses: both of its legs must find a place, and we then obtain knowledge through the relation between these two placings.

When the treatise is said to deal with “relation” and not with “relativity”, it is because the expression “relativity” is popularly apt to be misunderstood, in that one supposes (as in their way the old Sceptics did) that when something is relative, it means at bottom that it is invalid or indifferent, has no objective significance. In opposition to this misunderstanding it must be maintained that all truth finds its expression in the demonstration of quite definite relations and holds only within them, but then also holds absolutely there (provided, that is, that the relations are rightly determined). That all necessity, as Kant already taught, is hypothetical, and thus holds in relation to its presuppositions, does not mean that such necessity should not be real necessity. In all necessity the presuppositions must be asked after, for necessity consists in a relation between ground and consequent. Such a relation comes about only through the activity of thought. And from this point of view too the expression “relativity” is unfortunate, in awakening the notion of a resting property or state. But relation consists in referring, in a definite setting of thought’s two compass-legs. There lies in the concept of relation a direction to the work of thought.

2. The Concept of Relation in Psychology

Every attempt to give even a merely provisional explanation of what is meant by conscious life or mental life will have to lay particular weight on the fact that everything that is sensed, presented, felt and willed exhibits certain definite relations without which it cannot be thought. Attempts have certainly been made to conceive mental life as a chaos of relationless elements; but such attempts have not, in spite of the energy that sometimes lies at the basis of them, been able to lead to an intelligible psychology. The point is that from its first arising every mental element is woven into a connexion by which it is determined, and through which it can itself exercise a determining influence in mental life. It is hereby that mental life constitutes a whole, so that not least here the above-mentioned connexion between the concepts of totality and relation asserts itself. In such phenomena as memory, comparison, wonder and preference, the character of conscious life plainly displays itself, and in them there is precisely more than a chaos, in that the various elements which psychological analysis can distinguish in them are throughout determined in their peculiarity by their mutual relations. And all experience points to more primitive and simple mental phenomena having to be conceived on the analogy of these; in any case language has not formed special designations that could express the difference. We shall come to see in what follows that this hangs together with the degree of consciousness that can be present with respect to the relations of conscious life itself.

It is the thoroughgoing significance of the concept of relation in conscious life that makes it justifiable, as Leibniz and Kant especially have impressed, to characterize consciousness as synthesis. A relation can assert itself only when its terms are comprehended together; the relation is, indeed, precisely the expression of a connexion, of the terms' belonging to a whole.

The already mentioned double character of relation, as limiting and as carrying further, appears plainly in psychology. The peculiarity of every mental element rests in part on its relation to other mental elements, and is to that extent limited by this relation. But precisely by being determined through relations, every element, when it appears suddenly and without clear connexion with other elements, points beyond

itself and gives rise to a need for supplementing.¹³

As regards particulars, I must refer to my presentation of psychology, which is built precisely on the fundamental thought that the validity of the law of relation for mental phenomena establishes the character of consciousness as synthesis. Only a few principal features shall be recalled here.

As regards sense-sensations, it long remained undecided how far the concept of relation, and with it the concept of synthesis, held for them. In spite of Hobbes' energetic assertion that to have a single unchanging sensation would be one with having no sensation at all, one was inclined to conceive sensations as a chaos which "the understanding" was first to order. In the course of the 19th century one came to see the matter otherwise. Chevreul's discovery of the effect of contrast showed that the quality of sensations rested on the relation to the quality of preceding or simultaneous sensations, and Fechner demonstrated an analogous relation for the intensity with which the individual sensations assert themselves. — One has been inclined to conceive it as an imperfection that we do not have what has been called an "absolute" sensation of anything, and even Helmholtz regarded successive contrast-effect as an illusion. But no "normal sensation" can be demonstrated whose quality would be the "right" one. And in any case such a normal sensation would itself be conditioned by certain relations, which one then for one reason or another preferred to other relations. — We stand already here, with these elementary phenomena, before something that will show itself to hold for the highest intellectual functions. —

If we pass from sensations to presentations (which in their simplest form are reproduced sensations), it shows itself that the condition for a certain definite presentation arising at a definite moment is the relation in which this presentation stands to other presentations. It has indeed been attempted to ascribe to every presentation a sort of impulse to self-preservation, so that, as soon as other presentations (or sensations) do not lay claim to the psychical energy, it would "of itself" step forth into consciousness. But even so there are definite relations — inhibition or non-inhibition — on which

¹³In my treatise on Plato's "Parmenides" (Vid. Selsk. filos. Meddelelser. 1920. Kap. 3) I have drawn attention to the fact that while Plato found it difficult to find a logical transition from one concept (an Idea) to another, he entertained as a psychologist no doubt that even a suddenly emerging and in the highest degree valuable experience had its definite place in the series of experiences, conditioned by a whole series of endeavours and experiences having gone before in a definite order.

the appearance of the presentation depends. And the very independence with which the individual presentation asserts itself is conditioned, as experience shows, by its relation to the whole remaining conscious life, and can be understood only through this relation. There will always be something unnatural, and finally impossible, in keeping a presentation isolated from everything else in consciousness. Each single presentation has a tendency to call forth other presentations according to definite laws. Here again the synthetic character of mental life reveals itself. Most psychologists are accordingly agreed that the so-called laws of association are different forms of whole-association, that is, of the need for the reproduction of the entire state of consciousness in which the individual presentation was once a member. —

In the domain of the life of feeling, the law of relation was discovered long before its significance for other sides of mental life was recognized. Most plainly the significance of relation appears in wonder, whether this be a peculiar feeling or a member of more complicated feelings. But in all pleasure and displeasure, joy and sorrow, the relation between the present state and preceding states is of decisive influence. The happiness that is felt rests not merely on the definite magnitude of the gain or the progress experienced, but also on the relation to the level of feeling that was already present beforehand. The more different relations that are decisive for a feeling, the more peculiar and rich that feeling will be. With every such richly formed feeling the question will be whether it can maintain itself in face of the new relations in which mental life may be placed. Thus for example when the humorist is met by tragic fates¹⁴, or when a hard-won consolation is to stand its test in face of new dispensations.¹⁵ —

As little as sensation, presentation or feeling could arise under wholly uniform inner and outer states, just as little could there be any willing. To will is to prefer, and thus presupposes that inner or outer differences assert themselves. Reflex movement and instinct already presuppose changes in the inner organic state or in the outer relations. Reaction presupposes action. This holds both for involuntary and for voluntary willing. There is choice of all degrees, but a relation of difference is always presupposed. All deliberation, the involuntary as well as the voluntary, aims only at bringing such relations of difference out as plainly as possible. The various possibilities

¹⁴*Den store Humor*, p. 102–104.

¹⁵*Oplevelse og Tydning*, p. 150–152.

of action are measured by their relation to the momentary or hitherto existing state of mental life. There may here be a struggle whose outcome rests on the relations to the individual's character and past. One has sometimes conceived motives as absolutely independent elements, independent of the will itself. But every motive is itself a preferring, a willing. What can become a motive for an individual rests on the individual's character and previous history. Motives are never relationless. It is of decisive significance at what point of an individual's development a possibility of action appears; on that will depend whether a certain motive can come about at all, and whether it can become the victorious one.

But it is one thing that a proportionality asserts itself everywhere in consciousness, and another how far consciousness itself becomes aware of the relations that are decisive in the individual cases. Many psychological problems arise from one's not discovering the relations, these being perhaps discoverable only by objective, historical investigation — an investigation the individual is not himself in a position to undertake, at any rate not at the time of the experience itself.

Relations of contrast assert themselves from the first, both as regards qualities and intensities, without our noticing it. We are taken up with the life and the strength with which the new element brought out by the contrast appears. Thus when a colour stands "saturated", we readily forget the relation by which the saturation is conditioned. We turn towards a new feeling that comes forward with life and fullness after a feeling of the opposite kind, and forget the vanquished for the victor. An intention or a decisive resolution may announce itself as "a blow on the head", in that we forget the involuntary or voluntary deliberation that has paved the way for it through an interplay of memories, feelings and impulses. We do not always notice the course of association to which a presentation owes its coming into being; it may then stand as a sudden, unmotivated inspiration.

We have on the whole an inclination to dwell on certain salient points in our mental life and to overlook the relations that condition the transition from one such point to another. William James, who has excellently illuminated this peculiarity of mental life, distinguishes between resting-places and flying-places within the stream of consciousness, and holds that we have an inclination to keep to the resting-places.

He expresses himself more plainly when he speaks of substantive and transitive parts of our inner life, and holds that we are inclined to ascribe to the former an absolutely independent subsistence, in that the transitions are not easy to follow and often cannot be directly observed at all, but must be inferred to have been there.¹⁶ Such substantive elements are found precisely in contrast-determined qualities or intensities, in immediate recognition (the quality of recognition), and in a suddenly emerging presentation, a marked feeling, a surprising decision.

James has besides drawn attention to the fact that relations are sometimes felt as qualities, thus in those mental states which are expressed in language by “and”, “but”, “in spite of” and the like. It is a kind of sensation (or feeling) of connexion or of opposition, which can be just as immediate as the ordinarily so-called sensations. The so-called quality of recognition is of the same kind.

Here may also be reckoned the remarkable sensation of direction in our mental life which we can have. To use the comparison with the compasses, we may say that while the one compass-leg stands fixed (determined by our past and its results), we can feel that the other compass-leg is extended in a certain direction, without our being clear at what definite place it will come to stand. It is now more an indefinite expectation, now more an indefinite yearning, that here appears.

On the transition between psychology and epistemology lies the question whether, and how, a relation of which one has become more or less clearly conscious can be transferred to terms other than the original ones. An important way in which this can happen is the following. We involuntarily order our objects according to likeness and difference. Thereby series are formed, and it may then show itself that in such a series a term can be pushed out and the relation nonetheless continue to hold between its former predecessor in the series and its former successor. In the series ABC, perhaps B can be pushed out and the relation between A and C be the same as that between A and B.¹⁷ It is such a case that formal logic constructs as a relation between premisses and conclusion. The premisses become, for example: $A = B$ and $B = C$, and the conclusion becomes $A = C$. Relation — which precisely for that reason belongs among the fundamental categories — asserts itself generally prior

¹⁶*Principles of Psychology*. I, p. 243 ff.

¹⁷Cf. *Den menneskelige Tanke*, p. 162 f.

to all formation of concepts, judging and inferring. Already within sense-, memory- and fantasy-intuition, elements are ordered and reordered on account of their mutual relations, without there yet being any question of thinking in the narrower sense of the word. I refer here to my investigations of the relation between intuiting and judging in the first main section of “Den menneskelige Tanke” (especially p. 53–59). Of particular interest here is the way in which intuition is articulated quite involuntarily and unconsciously, without a judgment being formed. A striving asserts itself in conscious life to dwell in intuition as long as possible without forming concepts and judgments or drawing inferences.

In the series ABC, A remained standing, but C could enter into the same relation to A as B had had. The question now arises whether one cannot move both compass-legs, so that they are set at new places, but the relation between them continues to be the same. In the simplest case such a shift is present when there is a relation between two new terms (X and Y), and this relation is then recognized as the same as the one already known between two earlier terms (A and B). Such recognition can — like the recognition of qualities — be either immediate or mediate; in the latter case it presupposes explicit measurement, or else there are several relations serving as intermediate links between the two relations that are identified. The intuition in which, according to the legend, it dawned on Newton that the relation between the falling apple and the earth could be the same as the relation between the planets and the sun, was at first immediate; but it became mediate through the measurements and calculations that led to the definitive setting up of his doctrine of gravitation. On the whole many questions and problems arise precisely from the possibility of identities of relation appearing in one way or another before consciousness. Both questions of determination, which are due to lacking relations, and questions of decision, which are due to conflict between different possible relations, can lead thinking further. Already within ordinary consciousness (*sens commun*) such questions, with attempts at answering them, can appear, and what we call science rests only on the strict grounding and formulation of the questions and sharp investigation of whether the answer really is a solution.

It is an increasing degree of explicit consciousness of relations that leads over from

involuntary thinking to logical and mathematical thinking, which handles relations as simple qualities that can become terms in new series, and these again in series of series. An example of the transition is shown by the development of the concept of number from quality of number through ordinal number to cardinal number, and thence again to the various concepts of function.¹⁸

A definite limit for thinking would be present if there were series whose terms could be interchanged without further ado, because the relation of likeness or difference was absolutely the same between all the terms. Such series would be either chaotic series of difference or absolute series of identity. Whether they occur in existence cannot be decided in advance. But from the standpoint of psychology it must be said that they do not occur in actual mental life, where at most approximations to them can be found. It is a fundamental law in psychology that the order of the addenda is not indifferent. Where there seems to be chaos or absolute identity between psychical elements, it will be the psychologist's task to find, in the first case, likenesses or definite relations of dependence between the apparently absolutely different elements, in the second case, differences and the relations of dependence following from them within the apparently absolutely alike. Of chaos and of absolute identity one can speak only from definite points of view, and in relation to them.¹⁹ Thus the principle of relation asserts itself here too. The work of thought presupposes a constant reciprocal relation or a constant contention between continuity and discontinuity, likeness and difference; only through such contention does thought obtain its definite tasks and can begin to work. Hume's assertion of chaotic elements as what is properly given meant that on his conception no work of thought could come to begin. Spinoza's assertion of the eternal attributes as absolute identities presupposed that all work of thought was done. The two thinkers ended, each in his own way, in relations that really were not relations at all.

¹⁸*Den menneskelige Tanke*, p. 189–193.

¹⁹Anf. Skr. p. 165; 170.

3. The Concept of Relation in Epistemology

a. The Concept of Relation and the Other Fundamental Categories

The analysis that is to lead to the discovery of the forms and presuppositions of thought must be directed not merely at scientific work in the stricter sense, but also at the involuntary thinking we all practise in daily life, that which is characteristic of what one calls “sound common sense”. Out of this involuntary thinking, scientific thinking, which proceeds from definite concepts and seeks exact grounding, has developed step by step; and in the transition to science there cannot, however great and striking the contrast may be, be any complete change in the manner of thinking. Kant saw this when he stated that even “der gemeine Verstand” has certain “a priori cognitions” (*Kritik der reinen Vernunft*² p. 3), and shows it still more definitely in his doctrine of synthesis as the fundamental form of consciousness, springing from “a function without which we should not have knowledge, though we only seldom become conscious of it” (*Kritik der reinen Vernunft*¹ p. 78). We have seen above what significance this concept has for psychology. Like the other fundamental categories, this concept is one in which psychology and epistemology meet. In the most recent time Meyerson²⁰ has strongly and in a very instructive way emphasized that it is quite the same thinking that is practised in natural science, philosophy and sound common sense (*sens commun*). He does not, however, go more closely into the stages and paths by which practical thinking successively approaches science. He thus does not emphasize the significance of doubt as disquieting, in that practical orientation in the world does not always suffice, or else leads to conflicting possibilities in face of which one stands perplexed. It is in the school of doubt that thought comes to make stricter demands upon itself.

If one wished to define the word “reason” in such a way that it can fit pre-scientific as well as scientific thinking, one might say that it is a sum of categories, particularly of fundamental categories, which must be applied at every stage. Among these fundamental categories are to be reckoned the three pairs of concepts synthesis-

²⁰*De l'explication dans les sciences*. 1921. I p. IX–XIV.

relation, continuity-discontinuity and likeness-difference.

Relation is (like synthesis) a fundamental form which constantly recurs in all special forms of thought. It cannot then be surprising that it asserts itself particularly within each of the named pairs of concepts. Thus there is relation between synthesis and relation. These two concepts can, namely, come into opposition to one another, so that comprehending together and unity come to prevail at the expense of determination through relations, or, conversely, the terms of the relation assert themselves so independently that they do not seem to be members of the same comprehended whole. On the other hand, the need for synthesis becomes stronger the stronger the mutual tension of the terms is. When Kant says that we only seldom become conscious of the function that lies at the basis of synthesis, it is because it is only when the relation assumes the character of opposition, discord and tension that the comprehending work can properly come to consciousness. Herein lies precisely the significance of doubt. And the constant correlation between synthesis and relation shows itself in this, that doubt and problems exist only so long as different objects, in spite of ever so sharp an opposition or contradiction between them, are held together in explicit relation to one another. A problem falls away when one lets one of the conflicting terms fall away, or when one weakens the opposition between them.

There can arise a fear of the manifoldness and sharpness of relations, which makes the secure repose in involuntary syntheses appear particularly alluring. Thereby such a reaction is called forth as Hamann, Herder and Jacobi exercised in face of Kant's critical philosophy, and as James and Bergson signify in face of the science of the present day. And when Tagore calls upon Western Europeans to take up Indian ways of thought, there lies in this precisely a call to disregard all the sharp and tense relations that have been developed in European civilization, and to return to a relationless state. But however great and deep the search for unity may be that distinguishes the Vedanta doctrine, of which Tagore is the modern prophet, there is no way back. Either the relations that assert themselves are artificial and unfounded, and then criticism will overthrow them, or else they are founded in actual relations, and then they can be mastered only by a spiritual labour. Doubt is mastered only either by its groundlessness being demonstrated, or by the solution of a problem, or by the

demonstration that here an insoluble boundary question is present. When Descartes infers precisely from the possibility of doubt to the existence of thought, this meant a recognition of the close connexion between the categories synthesis and relation.

The relation between continuity and discontinuity is analogous to that between synthesis and relation, or is a more special form of it. Synthesis proceeds the more easily the fewer interruptions occur. If possible, such interruptions are involuntarily taken up into the intuition (in that this is more definitely articulated) or into the judgment (in that its terms receive additions that limit or extend). Where continuity prevails, intuition and reflection take on the character of a smoothly gliding stream without eddies and waterfalls. One does not ask, since no arrest or wonder motivates any question. Or, if one asks, the question runs: Why not? Why not go on as hitherto, when there is energy enough for the expansion to be continued? But in life and in science more or less sharp divisions of labour assert themselves, and thereby resistance and opposition which one cannot get round without further ado. One must then attempt to find intermediate links between the terms that have come to stand isolated in relation to one another. Or one must attempt to demonstrate a continuity lying deeper than the discontinuity that has forced itself forward, a continuity within which the leaps or breaks before us can come to stand as derived in relation to the law or the type the new continuity exhibits. From the purely psychological side this has already been mentioned above; it is on the whole the task of psychology to bridge the often so sharp oppositions and catastrophes that mental life can exhibit to immediate observation. Particularly of interest here is James' demonstration of "transitive" states, in contrast to the "substantive" ones. Psychological discontinuity can on the whole signify only a problem, never denote a solution. Analogously, in biology the doctrine of mutation, which asserts the sudden arising of new organic species independently of outer relations, signifies new problems. The founder of the doctrine of mutation himself, Hugo de Vries, at once stated that the mutations signify just so many problems. The modern quantum doctrine in physics, according to which energy is not released in a continuous way but in leaps²¹, is sought to be more closely grounded through the demonstration of mathematical laws according to which the release in leaps stands as

²¹POINCARÉ: *L'hypothèse des Quanta*. (Dernières Pensées. Chap. 6).

necessary. In every case theoretical points of view are sought for the understanding of the discontinuity. Already when the leaps can be ordered in a certain series, thought begins its work of finding a new continuity, even if this is for the time being more formal than real.

The problems that lie here were already raised in the earliest Greek philosophy. Parmenides' "One" meant at one and the same time continuity and identity; these were not different for him. Discontinuity, on the other hand, was asserted by Heraclitus and Democritus. Through Platonism and its opponents the dispute extends right up into more recent times. Kant regarded this dispute as a rational necessity. His doctrine of the antinomies aims at demonstrating here an absolute limit for reason. The great founder of critical philosophy did not see that discontinuity can never be thought's last word except in the form of a last problem, and his "antitheses", which assert precisely this point of view, are therefore decidedly right as against his "theses", which assert absolute discontinuities.

Continuity and discontinuity are correlata that reciprocally supplement one another. They denote different points of view and operations, and the history of science shows how now the one, now the other of these categories comes to stand in the first rank, but in such a way that the struggle between them will always begin anew. No investigator has illuminated this relation better than Henri Poincaré, especially in the following statement: "This struggle will last as long as men will cultivate science, as long as humanity will think, for it springs from two irreconcilable tendencies in the human mind, two tendencies which it cannot lose without ceasing to exist, namely the need to comprehend, and we comprehend only the limited, and the need to intuit, and we can intuit only extension, which is unlimited. Even if this struggle should not end with the final victory of one of the combatants, it would not for that reason be fruitless, for in each new struggle the field of battle is another; each time, therefore, a step forward is taken and a conquest is made, not for one of the combatants, but for humanity."²² — To this fine and apt statement I must, however, add the remark that

²² *Les conceptions nouvelles de la matière* (in the collective work "Le Matérialisme actuel", 1913), p. 67. — Cf. also the interesting discussion in the French philosophical society 13 March 1910 (Bulletin de la société française de philosophie. Vol. X), especially the statements of Perrin, Brunschwig and Meyerson (p. 116–120).

the relation between intuiting and thinking can also have another character than that indicated by Poincaré. It is often precisely intuition that holds fast to limited images, while thought, in virtue of the recognition of law, sees the limitation of every intuition and the possibility of new terms in the series of images. Besides, there are several transitional forms between intuiting and thinking, which have in part been indicated in the foregoing, and there will in each individual case be definite objects, given in experience, which decide whether intuiting or thinking shall have the word.

Where continuity can be maintained, the condition will be that no great differences in time, place, intensity, magnitude or quality assert themselves between the objects with which consciousness is occupied. Where likeness prevails, one passes easily and, at any rate apparently, continuously from the one object to the other. Likeness and difference are a pair of concepts exhibiting an analogy with the pair continuity and discontinuity. But while it can be said of continuity that it is not noticed until it is past, likeness has a somewhat more conscious character. It does not stand in the close relation to intuiting that continuity does. — The greater the likeness, the better will continuity be able to be preserved. It is only through explicit, perhaps voluntary comparison that one discovers the difference between the various degrees or species of likeness. I have (in my *Psychology* and in “*Den menneskelige Tanke*” p. 64 f.) distinguished five species of likeness: identity, congruence, composite likeness, likeness of quality and likeness of relation (analogy). In this series the element of difference comes forward more and more, and it is essentially through increasing difference or decreasing likeness that we pass over from continuity to discontinuity.

The relations likeness and difference are of such essential significance that it is justifiable to define thinking as comparison. That the two relations govern all thinking and all categories was already seen by Porphyry, when he found it necessary to write an introduction to the Aristotelian treatise on the categories²³, in which he explained the concepts genus, difference, species, individual and property. It was this “Introduction” that came to lie at the basis of the many discussions in the Middle Ages about the objective validity of general concepts (the dispute between nominalists and realists). The five concepts cited form a progressive series as regards differences, thus

²³*Isagoge in categorias* (Scholia in Aristotelem ed. Brandis, p. 1).

from the abstract or general to the concrete or special. For the thinking of the Middle Ages, which was predominantly classificatory and syllogistic, the question which of these concepts was to be applied in the individual case, and with what right it was applied, was of the greatest significance. More recent science here undertakes a great reduction. For it, it is not the relation between higher and lower concepts that is decisive, but everything comes down to whether a concept (an object) can step into the place of another in all respects or not. It is identity that puts the other species of likeness in the shade. And precisely by the fundamental character of the principle of identity as the nerve of all inference, and by the sharp determination of the concept of identity conditioned thereby, modern science distinguishes itself from ordinary consciousness, which does not take the difference between identity and mere likeness so exactly. Parmenides had, moreover, already seen that the nerve of thought lies here, though it was first Leibniz who laid the concept of identity at the basis of logic, because it made substitution possible. It is in this connexion indifferent whether one defines identity as the greatest possible likeness or as the least possible difference; both definitions testify to the correlative relation between the categories likeness and difference. — But at the decisive turning point denoted by the dominion of the concept of identity, we pass over from the fundamental to the formal categories.

b. The Concept of Relation and the Formal Categories

α. Through the working out of the formal categories (identity, analogy, negation, rationality) it becomes possible to obtain exactly determined relations.

The concept of identity is formed as a natural continuation of the tendency which practical thinking already has to form series of objects ordered according to increasing likenesses or decreasing differences. Its formation makes possible points of view from which not merely an overview of the world of relations can be won, but by whose help the one relation may perhaps be derived from the other. Only at this point do the epistemological and the psychological-historical ways of consideration come into decided opposition to one another. Psychologically and historically, at most approximations to identity assert themselves, and the pure concept of identity is perhaps even rejected, in order that the differences and changes which observation

exhibits may come into their full right. Epistemologically the relation is considered from the opposite side: differences and changes stand as breaches of the identity which is the first presupposition of all science, and which conditions what becomes a scientific problem.

The fundamental significance of the concept of identity for scientific thinking rests on the fact that its application is the presupposition of the two most important functions of thought: classification and proof.

Division of the various objects which observation shows us can become of decisive significance only when there is a point of view from which the objects, in spite of all difference, can be regarded as identical. In concepts of genus and species such points of view are given. The individual objects for which such concepts hold can — seen from the point of view of the concept in question — be regarded as one and the same object, which displays itself in the fact that any one of them can be used as an example of the concept. The differences between the individual objects do not thereby fall away, but can, precisely by being united under the same point of view, come forward as sharp oppositions. There is perhaps not even a single one of those properties which the concept states to be “common” that occurs in quite the same way in the individual objects exemplifying the concept. But an analogy nonetheless prevails between the various objects, in that the relation between their properties is throughout the same.

Aristotle laid it down with great emphasis, both that science concerns only the general, the content of concepts of genus and species, and that what really exists is always individual. Of the individual one could in his opinion form no concept. He has thereby, without being clearly conscious of it, posed a great problem which extends under various forms through the whole later history of scientific thinking. Considering the question here provisionally only from the point of view of classification, it must be pointed out that concepts can nonetheless be formed whose content is those properties or relations which, in spite of all differences in an individual object’s several states, can express the object’s identity with itself. Such typical individual concepts there will be possibility and reason to form more in the domain of the human sciences than in that of natural science. Between the various states of the same object there prevails, as between the various examples of the same general concept, only analogy, not identity.

Where neither analogy nor identity can any longer be demonstrated, we stand before a chaos. But the concept of chaos itself acquires real significance only through the relation of opposition to identity. There is a difference between the chaos that stands before ordinary consciousness when it is overwhelmed by the differences, and the chaos that is posited by a thought which in serious labour has searched for analogies and identities, and now at last lays down the work. And even in this latter case it can never be maintained with certainty that we stand before a chaotic series of difference. Strictly speaking it is not justifiable to say anything more than that it has hitherto not proved possible to find points of view of unity for the objects that seem to form such series. Just as identity can be defined by the greatest possible likeness or the least possible difference, so chaos can be defined as the greatest possible difference or the least possible likeness. The likeness is so small that it has hitherto not been noticed, just as with the objects we regard as identical we can say that the differences are so small that they are not noticed.

The concept of negation emerges here as an expression of the fact that an interruption of the work of thought — here of the dividing work of thought — shows itself necessary, since no further points of likeness can be found. The history of classifications shows that in such cases one often forms negative concepts (or concepts containing only negative marks), as when Lamarck divided animals into vertebrates and invertebrates; later Cuvier found positive marks for the invertebrates and the possibility of a positive division of them.

The chaotic series of difference as well as the absolute series of identity denote two ideal boundary lines for knowledge, boundary lines which can never be demonstrated in actual knowledge. The one of these boundaries would denote the impossibility of knowledge²⁴, the other its conclusion. Between the two lies a scale of series of thought (some of the most important of which are presented in “Den menneskelige

²⁴Stuart Mill attempted to lead all inference back to an associative transition from one particular to a particular different from it. But such a transition is, even purely psychologically, possible only when the first particular is recognized; only then can it be understood how it can call forth the second particular with which it was formerly conjoined. It is, however, rather an inference from analogy that makes it possible to infer “from particular to particular”: as a in an earlier observation stood to b, so a' will now stand to b'. But then there must after all be likeness between a and a'. Cf. *Den nyere Filosofis Historie* (First ed. II, p. 377; Second ed. II, p. 414). (In the third edition: III, p. 119). Cf. in the most recent time: ETTORE GALLI: *Nel regno del conoscere*. (1919) p. 159—167.

Tanke” p. 162–174), denoting different stages by which knowledge works its way towards its ideal. —

So long as knowledge aims only at a classification, it lets the differences subsist and merely seeks to attain an ordered overview of them. In proof, on the other hand, the concept of identity acquires active or productive significance, affords a means of demonstrating new truths, perhaps of finding new objects without constantly relying on observation. Here the category of rationality emerges, which appears in the relation between ground and consequent. Its presupposition is an identity between two objects justifying us in everywhere putting the one in the place of the other. The two premisses from which the inference proceeds must contain a concept that is identical in them both. Classification already makes inference possible, after identical concepts of genus and species (or typical individual concepts) have been formed. But science seeks to go a step further. It forms not merely ordered series of known terms, which can be continued only by the help of new observations, but also series whose continuation can take place according to the same law by which it began. Here the difference between logic and mathematics appears. The mathematician sets up not merely the principle of identity as holding for the units with which he works, thus $1 = 1$, but he sets up also the presupposition that out of these units he can form a series arising by unit being added to unit in always the same way, thus $1 + 1 + 1 \dots$. A presupposition such as this last is unknown to pure logic. For it, repetition of an object has no significance for thought, however much psychological or pedagogical significance such repetition may have. It holds to the principle of identity (and the logical principles more or less directly connected with it). The fundamental mathematical presupposition is, on the other hand (alongside the principle of identity), the positive and productive significance of repetition. It is this presupposition that Poincaré has called *loi de recurrence*, and in which he finds the significance of what Kant has called an a priori synthetic judgment.²⁵

While classification left the qualitative differences as they were, and merely ordered them, modern natural science seeks to work up or purge the differences of quality in such a way that out of them series of a kind similar to those just mentioned can

²⁵Cf. on this *Totalitet som Kategori* (Vid. Selsk. Skr. 1917)

be formed, thus purely formal series in which new terms can always be constructed according to the same law which has already given the preceding terms their place. There are four species of qualitative difference which have particularly made such a purging possible, namely time, number, degree and place. To immediate observation they appear not as “pure”, but as qualities. The difference between times, between places, between numbers and between degrees is just as much qualitative as the difference between yellow and blue, hardness and softness. The history of science shows what work of thought has been necessary before the four named series of quality could be transformed into series in which each single term is fully determined by its place in the series, a place itself determined by the relation to the other terms. What fills the moments or happens at the different places, what is counted or measured in degrees, is now indifferent. Each single term in the series is regarded as though it had the same content as the others, so that only the place makes a difference. In brief outline I have described this process of purging in “Den menneskelige Tanke” (p. 186–201), and I shall not go more closely into it here. I shall only add that the four processes of purging do not take place independently of one another, but reciprocally support one another, and that this becomes possible through the categories arising sporadically in general; a certain development of the one category can thereby become significant for the development of the other. Thus the concept of time has developed with the support of the concept of number and the concept of space. And it seems as though the concept of number has on the whole been of great help in the purging of the three other categories. The process by which unit is added to unit is simpler and easier than that by which moment is added to moment, degree to degree and place to place. Kant already saw that in counting there was the simplest example of “Synthesis des Gleichartigen”. Comte has emphasized this still more strongly, and Gauss too has stated that number is in a far higher degree than space a product of our own mind.²⁶

While typical individual concepts, which concern concrete objects, can be devel-

²⁶In the “Reflexionen” edited by Erdmann, KANT says that it is really only numbers and magnitudes that can be constructed out of pure reason, because with them mere repetition is needed. (Cf. *Totalitet som Kategori*, p. 36). — COMTE found in pure speculations about number “la véritable origine de tout le système scientifique” (*Discours sur l’esprit positif*. 1844, p. 100. — GASUS declared in a letter that number is a product of our mind, while space has also a reality outside it. (See the citations in E. MACH: *Erkenntnis und Irrtum*¹. p. 384).

oped only with constant regard to and borrowing from observation, there has through the named processes of purging been provided the possibility of forming series of thought which can be continued according to the same principle that lies at the basis of their formation, and from which demands can be made of the objects observation exhibits.

β. In the formation of the mentioned pure series of quality the concept of identity everywhere lies at the basis, in that they are characterized by the concepts time, number, degree and place being freed from all content that fills the time, constitutes the number, exhibits degree, occupies place; with the content all real differences within the series fall away. That they must still be called series of quality comes from the differences between time, number, degree and place among themselves continuing to subsist. There is, as there will later be occasion to mention, a tendency to let number become dominant at last and to make all knowledge a system of equations. Yet here, as Meyerson has shown²⁷, algebra and geometry contend for precedence. The question itself belongs under the real categories.

But however high a degree of exactness the knowledge working with the named series may attain, the concept of relation announces itself even in these pure regions. This shows itself especially at three points.

Identity is itself a relation, an object's or a thought's relation to itself under different circumstances. Whatever definition of identity one employs, determination through relation is not avoided. Whether one speaks of the greatest possible likeness or the least possible difference, the tinge of relation does not disappear. And this tinge becomes particularly strong when one employs Leibniz's experimental definition: identical are those thoughts or objects which without cancellation of validity (*salva veritate*) can everywhere be put in the place of one another.²⁸ By making everything

²⁷*De l'Explication dans les Sciences*. I p. 264; II p. 167; 204–220.

²⁸As mentioned in "Totalitet som Kategori" (p. 41 f.), various conceptions have asserted themselves among Danish mathematicians in face of the question whether their science requires fundamental propositions expressing absolute identity, or whether approximation to such is sufficient. This opposition recalls the two different definitions of identity — as greatest possible likeness or as least possible difference. In the last work we have from ZEUTHEN's hand, he states (*Hvorledes Matematikken i Tiden fra Platon til Euklid blev rationel Videnskab*. Vid. Selsk. Skr. 1917, p. 80) that the applicability of Euclid's geometrical system rests on *that* the empirical lines have the properties stated in the postulates, which Euclid presupposes, or if one will, it depends on *how far* they have them. By this "how far" there

depend on the possibility of substitution in all cases, one plainly enough employs precisely the relation between the thoughts (objects) in question and others; validity is preserved only if such relation is not altered when the substitution takes place. Even with the identity $1 = 1$ the question may arise whether we really dispose of units that can everywhere be put in the place of one another. In the discussion of Fechner's attempt to give Weber's law a strictly mathematical formulation this question came forward.

But secondly, the setting up and application of the concept of identity always presupposes a definite point of view from which alone it holds. In classification it is plain enough that the identity holds only for certain marks which constantly recur, so that there is at least analogy between their forms in the individual cases. But the identity which is the nerve of every proof also holds only from a certain point of view. Identity is always attained through purging, that is, by disregarding points of view that do not concern the matter under discussion. When the various forces of nature are determined as energies, that is, measured by the capacity to perform work, there is thereby provided a point of view of unity, just as when otherwise highly different things are considered solely and alone under the point of view of weight. Right up to the appearance of spectral analysis one regarded, in accordance with Newton's physics, the globes in space essentially under the point of view of weight. It is by adopting a purely quantitative point of view — a point of view which again becomes possible only by one's holding to certain definite properties — that identity can be provided between otherwise highly heterogeneous objects. Just as in the descriptive sciences quality can be reduced to identity by means of concepts of genus and species, so in the exact sciences quality can be reduced to identity by means of quantity.²⁹ Plato has given profound hints in both directions; as regards the last point, Galileo took up his thought in an ingenious way.

In Greek thinking a kind of cult was often practised towards the "One", the sum of the pure identities. It appears in Parmenides and Plato and reaches its high point in

is surely an allusion to the conception which HJELMSLEV has asserted, and according to which ideal definitions such as the Euclidean are not needed, but one can proceed from lines and figures that can be drawn.

²⁹Cf. *Den menneskelige Tanke*, p. 177–180.

Neoplatonism. For the thinking of the more recent age, identity is not an object of contemplation, but is a great means of thought, as is aptly expressed in what I have above called Leibniz's experimental definition of identity. And not less essential is it that we go a step further back and ask from what point of view the identities hold. We do not on that account miss a world of Ideas. On the contrary, the points of view from which identities become possible are for us the true world of Ideas; and it is a consideration which epistemology contributes to the discussion of the problem of existence (the cosmological problem), that the fact that there are such points of view, by which the strictest knowledge men can attain is possible, must necessarily become of great significance for a final, rounding-off world-view. —

Thirdly, it is precisely the formation of the qualitative series of identity that makes the relations come forward more sharply than is the case before the purging. In the named series each term is determined exclusively by its place in relation to the other terms. A moment acquires, quite apart from its content, an individual character by following after certain moments and preceding certain other moments. The moments serve reciprocally to measure out one another's individuality. The same holds of number: each number in the series is an individuality, precisely in expressing a definite relation to other numbers. Malebranche already remarks that whole numbers are relations just as much as fractions are, although one does not always notice it, because whole numbers can be expressed by a single sign. As Gauss later stated: "Every real whole number represents the relation of an arbitrarily chosen initial term to a definite number in the series".³⁰ The cult of certain numbers which is so frequent in popular religions became possible only by numbers being taken out of the series in which alone they have their individuality. — What holds of moments and numbers holds analogously of degrees and places. This needs no closer demonstration in this connexion. As regards place (space), another, more special connexion will give occasion to return to the question. Of interest, however, precisely here is Poincaré's

³⁰MALEBRANCHE: *Recherche de la vérité*. VI, 1, 5. — GAUSS cited in CASSIRER: *Substanz und Funktionsbegriff*. p. 72. — This is, however, expressly denied by BERTRAND RUSSELL (*Introduction to Mathematical Philosophy*. p. 64), for whom the number m is no relation; it is otherwise with $+ m$ and $\div m$. A number is for Russell's Platonizing conception an existence in and for itself, apart from the relations in which it is seen. Therefore Russell also distinguishes definitely between $\frac{m}{1}$, which is a relation, and m itself.

remark³¹ that the relativity of space and its uniformity are one and the same thing. —

γ. To the formal categories we have besides identity and analogy also reckoned negation and rationality. Nor in their case is any detailed demonstration needed that relation constantly lies at the basis. — Every No presupposes a (possible or actual) Yes. A negation is purely logically a sign of a path that cannot be travelled, and thus presupposes attempts to go such a path. Or it presupposes a comparison between two intuitive images, of which the one has elements lacking in the other. Mathematically, negation can signify a call to form a series in the opposite direction from the earlier one. — Rationality signifies in itself a relation, a relation between premisses and conclusion. And as far as the demand for grounding goes, so far must the presuppositions of our judgments be asked after. An unconditional necessity does not exist. Necessity expresses only the relation between ground and consequent, a relation that must constantly recur so long as there is thinking, whether it be possible to find it in the individual case or not. Just as in classification we came upon the concept of analogy, in that, where concepts of species can no longer be found, we stop at analogies between the individual objects (in so far as such analogies can still be found at all), so, as regards proof, we stop at the so-called inference from analogy. Aristotle has aptly called it inference from an example. From the relations within one object we are led to surmise corresponding relations within another object. The more examples we hereby have to proceed from, the more does the inference from analogy approach induction. But it always acquires (like induction, for that matter) significance only as a method of discovery, that is, as an occasion for setting up propositions which can thereafter be more closely tested; it can never become proof. Therefore it has aptly been called a good servant but a bad master. That analogy too always rests on definite presuppositions needs no closer demonstration.

c. **The Concept of Relation and the Real Categories**

α. The question naturally arises whether it is possible to apply the series of which the formal categories show the possibility to objects given to us quite apart from our own constructions; and that again is to ask whether out of such objects series can be formed

³¹*Science et Méthode*, p. 113.

which in some degree or in some way correspond to those formal series and can thereby share in the necessity distinguishing the latter — thus whether a rationalization of given objects is possible. The relation between ground and consequent is, after all, the model of all necessity.

This question gives occasion for a discussion of the real categories, namely causality, totality and development, which I have presented in “Den menneskelige Tanke” p. 207–237, and here return to only in order to see what part the concept of relation plays in their domain.

The real categories have historically not waited for the formal ones. Rather it is they that have given the impulse to the working out of the latter. The development of science out of practical life and sound common sense appears strikingly here. The real categories, particularly the concept of cause, have their actual origin in that need for orientation in existence which is a condition of life for animals and for men. The development of thought out of doubt shows itself strikingly here. When disappointment, resistance or contradiction puts an end to the involuntary trust in all emerging sensations and presentations, there is no other way to go than to seek new sensations and presentations which can exhibit a firmer connexion than the one that was broken, and which can make the secure continuation of life possible. But this means that the circle of relations is widened until the various objects show themselves to stand in fixed, unavoidable relations of connexion to one another. It is causality or lawful connexion that lies at the basis of all conviction of reality which is possible after the involuntary trust in the immediately given connexion is broken. This practical criterion of reality is deepened and made precise in science through closer testing and determination of the relations conditioning the connexion between the objects. Beyond all relations thought does not get. Of the reality of something that should stand in no relation at all, no grounded conviction can be possible. Reality and relation belong, after all, together. Every establishment that something is real consists in the demonstration of a fixed relation between this something and the circle of objects which hitherto stand as real through mutual fixed connexion; the one compass-leg of thought stands in this circle, and with the other compass-leg the relation between this circle and that whose reality is in question is measured out. If doubt is raised about the

reality to which one has hitherto held, then the first compass-leg too must be moved, and so on. There will constantly be use for both compass-legs in the knowledge of reality. If one is forced to stand on the one without being able to find a secure place for the other, one is in a state of doubt, expectation or longing.³²

On account of the close connexion between causality and the criterion of reality, which shows itself in our having grounded conviction of reality only where we can connect objects as cause and effect, it was a revolutionary step that Hume took in doubting the right to apply the concept of cause. He shook both the scientific knowledge of existence and the practical criterion of reality. But his criticism rested on the presupposition that cause and effect were two wholly different things — terms in a chaotic series of difference. Here precisely the fixed connexion is lacking, since one can place the terms of such a series in any order whatever. Now it is a fact that we nonetheless sometimes stand before series whose terms cannot be interchanged, and it was in this fact that Kant took his point of departure as the basis at once for the principle of causality and for the concept of reality. And the history of science shows how the external relation between cause and effect in many domains gives place, in the course of investigation, more and more to a relation of continuity between the objects standing in causal relation to one another, and this again perhaps to a relation of rationality, in that later terms in the series of events can be derived from the law of the sequence of the preceding terms.

However continuous and rational the series formed in this way may become, the concept of reality nonetheless continues to stand as an ideal which can never be fully covered by observation. New intermediate links will always be findable, and new objects may emerge whose continuous and rational relation to the earlier ones is to be demonstrated. Reality constantly means a relation to continuous and rational series

³²Thomas Aquinas attempted to free the theological concept of God from all relation by maintaining that the world indeed stands in relation to God, but God not to the world. (Cf. on this my *Religionsfilosofi* § 22). Thomas here attached himself to mysticism, particularly to Hugo a St. Victore, who again is influenced by Neoplatonism. Proceeding from this train of thought, a modern Catholic theologian has criticized Spinoza for not having dared to take the step of assuming that to the world's dependence on God there corresponds no dependence of God on the world. (DUNIN-BORKOVSKI: *Der junge Spinoza*, p. 352). But a relation always presupposes two terms, and however one may stand towards Spinoza's concept of God, one must acknowledge the consistency with which he holds fast to the concept of relation on this point. Cf. my treatise *Spinoza's Ethica*, p. 26; 32–35; but also p. 96 f).

formed hitherto. And no guarantee can ever be given that these series may not have to be replaced by quite other series, or, to keep to an image used above, that both compass-legs may not have to be moved.

Herbert Bradley has rightly employed the concept of experience in the for us inconcludable characterization of existence. Experience is for Bradley the measure of reality; but since experience is constantly won by the demonstration of relations, and new relations can always assert themselves, a complete satisfaction of that measure's demand is impossible. Concluded harmony and relation are irreconcilable concepts. Bradley is at one and the same time convinced that we always think in relations, and that a complete solution of the problem of reality is impossible from the concept of relation (the relational point of view).³³

The solution can, and in this Bradley remains right, never be fully attained. But the work of knowledge constantly aims at finding new relations and seeking to bring them into connexion with relations found earlier, perhaps using them to correct the latter. Without here entering upon what belongs under the cosmological problem, it must be maintained that the work of knowledge is itself a reality, a member of the great reality which is perhaps far more comprehensive than what any work of thought can span. And when that part of reality which the work of knowledge constitutes can never be concluded, one might perhaps draw from this the surmise that existence on the whole is not finished and cannot become finished. Such a surmise³⁴ presupposes, to be sure, that the form of time holds for existence, not merely for our thoughts, although at all times, also now, there are investigators who do not assume this. As regards Bradley, he definitely denies that thought is a part of reality: The process which moves within Reality, is not Reality itself. Philosophy is itself but appearance.³⁵

Bradley's profound principal work is cosmology ("metaphysics"), not properly epistemology. He sets reality in opposition to truth: when truth becomes complete, it ceases to be truth, for then reality enters in its purity. From this standpoint it follows

³³*Appearance and Reality*¹, p. 171 (When thought begins to be more than relational, it ceases to be mere thinking); cfr. p. 445; 519.

³⁴Such a surmise I already stated in my University Programme of 1902 (*Filosofiske Problemer*) p. 61 and shortly afterwards in lectures in America (*Mindre Arbejder. Tredie Række: En filosofisk Bekendelse*, p. 26 f.).

³⁵*Appearance and Reality*¹, p. 448; 454.

that although he lays such great weight on the concept of relation that he defines thinking as finding relations, he nonetheless conceives thought's search for relations as an imperfection, because conclusion and totality are thereby excluded. And with this it again hangs together that he misjudges the significance of the relations with which the special sciences each work in their own domain. He has clearly seen the significance of the concept of relation as basis and measure for all search after truth. But his own system is at bottom a crystallized ideal of truth, an attempt to think experience as complete (experience entire, containing all elements in harmony, p. 172), — an attempt which cannot possibly be carried through, precisely when one conceives thinking, as Bradley does, as relational.³⁶ —

The demonstration of continuous series of objects with non-interchangeable terms is not yet science's highest ideal. It provisionally remedies the external relation between what one calls cause and what one calls effect, which called forth Hume's scepticism; but it does not completely put an end to it. A step further is led by the demonstration of two-sided equivalence between different objects. There may then be the possibility both of deriving the one object from the other and this other from the first. Where the relation of equivalence is formulated as an equation, this appears plainly. It is (in theory) indifferent from which side of the equals sign one begins, and the qualitative difference, as well as the temporal difference between the two objects, then no longer has any significance; an absolute relation of identity is attained. The qualities (among which time, succession, may also be reckoned) then stand as mere "anthropomorphisms", subjective objects about which strict science does not trouble itself. One here goes a step further than the mechanical conception of nature, which also disregarded qualities in the narrower sense, but still acknowledged time as an objective form, in that it saw motion in space as the only thing that existed, although it was "falsified" by men's subjective apprehension.³⁷ It is a significant and paradoxical

³⁶In the same direction as the criticism of Bradley given here go HALDANE's critical remarks (*The Reign of Relativity*, 1921, p. 203—211) towards this thinker, to whom he stands in other respects so near. Haldane's work first came into my hands after the present treatise was completed.

³⁷THOMAS HOBBS, who partly by his own path, partly under the personal influence of Galileo, became one of the first champions of a strictly mechanical conception of nature, expresses in his poetic autobiography the insight he had won (motion as the only thing) thus:

Et mihi visa quidem est toto res unica Mundo

step that is taken further when one assumes that time can be reversed, so that the sequence of events is at last indifferent.

In opposition to such a conception, which is often maintained from the mathematical and natural-scientific side, and which in philosophical form has been carried through by the Marburg school, it must be maintained that the world of pure equations in which strict science ends, or towards which it at any rate aims, does not lose its significance because it is constantly held fast that a relation between before and after can never completely resolve into a relation of pure identity. Because there is a two-sided relation of equivalence between two qualitatively different objects (such as heat and motion), it is nonetheless not indifferent which object comes first and which follows after. Something different happens in the two cases, which will plainly show itself when the series is carried further through new terms: even if there is a relation of equivalence between A and B, there may from B be a transition possible to C which would not be possible from A. Besides, the transition does not in fact take place equally easily in both directions (as Carnot's theorem shows with respect to the relation between heat and motion). But in spite of everything, that strictly mathematical conclusion of the knowledge of nature has its great significance in this, that to the purely timeless, logical-mathematical relation between the terms of the equations there corresponds the temporal transition between qualitatively different objects in our experience. It is through analogy that logical-mathematical knowledge acquires its inestimable significance. In the equations the expert can read off the course of the phenomena, just as the musician can read off the course of the notes in the score.

This conception was already hinted at by Kant in his doctrine that between ground and consequent on the one side, cause and effect on the other, there existed only a relation of analogy, not a relation of identity. It has later been asserted in a particularly significant way by CLERK MAXWELL (in the second volume of his "Scientific Papers"). In my University Programme 1902 ("Filosofiske Problemer") and in "Den menneskelige Tanke" (1910) I have set forth the same conception, which I have in "Totalitet som Kategori" (1917) developed more closely in polemic against the Marburg school.

That the relation between a series of equations and a series of qualitative changes is not a logical relation of identity is seen from the fact that from pure logic and mathematics one could never derive that there is a temporal world with qualitative differences. It goes with that world of equations as with Plato's world of Ideas (which at bottom appears here in a new shape): one can perhaps climb up to it, but the ladder falls away as soon as one has got up there. And to this comes further that a conclusive proof of the real validity of the absolute series of identity cannot be given, since there is always the possibility of new relations whose analogies with the series of identity found must first be established. Here again it shows itself that the concept of relation makes truth and reality ideals which can be attained approximately only through constant labour.

ÉMILE MEYERSON has in his epistemological works laid very great weight on the difference between knowledge of a lawful relation between objects and a rational explanation of such a relation. Especially in the recently published great and penetrating work "*De l'Explication dans les Sciences*" (1921) he seeks to illuminate this difference through a wealth of examples from natural science and its history, a wealth which everyone who occupies himself with epistemology must envy him. For my part I agree with him on the main point, although my terminology is somewhat other than his. I have already, in the section of my *Psychology* that deals with the transition from psychology to epistemology, maintained that science does not, and cannot, stop at the mere establishment of an unavoidable and non-interchangeable sequence of two objects, but that the task, after such a sequence (and thereby a "law" in the popular sense of the word) has been demonstrated, becomes that of finding a continuous transition from the one term in the causal relation to the other. so that what we call the effect can come to stand as a continuation of what we call the cause, a continuation forming an analogy to the way in which in an inference the conclusion "follows" from the premisses. It is the great ideal which the thinkers of the 17th century set up when they demanded that there must not be more in the effect than there was in the cause. The continuity found when this ideal is to a certain degree attained is, so far as I understand, what Meyerson calls rationality, and what is attained by "explanation". In a very instructive example he shows (I p. 315) how science does

not stop at demonstrating a lawful relation between certain optical phenomena and the chemical structure of a substance, but that one seeks to find a connected process beginning with the action of light, which is a certain kind of motion, and continuing through the motions which this action calls forth in the interior of the molecules, and which again call forth certain motions in the sense organs affected by the substance in question. It is according to this scheme a continuous series of motions that is sought to be established. But in so far as it is possible to carry this scheme through (and Meyerson very strongly emphasizes the difficulties in it), it can be done only through the demonstration of a series of laws (for the reflection of light, for the inner structure of molecules, and so on), and I do not see what opposition there then finally remains between “lawfulness” (*légalité*) and “rationality”. Meyerson himself accordingly sees clearly that every knowledge of law, often unconsciously for the discoverer, presupposes an inner bond between the objects that are tied together in virtue of the law (II p. 289), and he concedes (II p. 337) that legality and rationality are in reality so closely (inextricably) connected that it is only by reflection on the foundation of science that one can keep the two principles apart. For my part I do not assume two principles here; it is one and the same principle carried through with increasing exactness and consistency.

I therefore do not believe either that Meyerson is quite right in his criticism of Comte, when he formulates this criticism by saying that Comte indeed acknowledges lawfulness, but not rationality, as the task of science. Comte himself states definitely (in *Discours sur l'esprit positif*, p. 20–21) that the task of science is both explanation and foresight, and that it “by its systematic speculations meets our understanding’s involuntary need for unity by carrying through the continuity and uniformity of our various concepts”. It is another matter that he did not think there was any prospect of realizing this ideal, partly because the actual discontinuities seemed to him insuperable, partly because he wished to be a founder of religion and to make the results of the science won hitherto (i.e. up to 1830) the dogmas of the new religion. The criticism of Comte must in my view be formulated thus: he demanded a conclusion of the knowledge of law and did not see the possibility of carrying it through in still more special forms and degrees than had hitherto been done. —

There is yet another point on which I do not agree with the excellent French thinker from whose works I have learned so much. I do not believe that science's last goal is the setting up of general propositions, formulated as exactly as possible. There is yet a work to be done, namely to use the general propositions, the rational points of view, for the understanding of the individual existences which subsist only by means of an interplay, a manifold conjunction of lawfulnesses. Aristotle was wrong that the individual could not be an object of science. Rather the matter stands thus, that the immense work science does to find lawfulness and rationality is itself again the condition for winning an understanding of the individual wholes which are what actually exists. Only by not forgetting this task is the eye opened for science's great, infinite goal.

β. This reminds us that besides causality, totality and development too must be reckoned among the real categories — precisely in virtue of the concept of relation. For the rational points of view that have been found must be connected with one another and are thereby placed in definite relations to one another, and through the work on this thought finds new totalities where formerly there seemed to be only sporadic observations. But every such totality stands in relation to objects and elements outside itself, and therefore — in virtue of the concept of relation — ever new tasks are set. When, on the other hand, given objects immediately appear as wholes, the task becomes not synthetic but analytic; the point is to find the connexion, the interplay, that makes such wholes possible. What one calls "things", immediately given wholes, are understood only through their properties, and the properties denote just so many relations to other "things". Molecules, atoms and electrons are constantly "things", wholes known and understood only through their relations.³⁸

The concept of development comes forward naturally when one lays weight on causality signifying a continuous process; different stages can then be distinguished here until a certain conclusion is present, and a characterization of these different stages can be attempted. But further, the concept of development hangs closely together with the concept of totality, in that every whole has its history. — What in the individual case is called a whole or a stage of development will depend on the point

³⁸A closer exposition of what is here hinted at is given in my treatise *Totalitet som Kategori*.

of view. What for one consideration stands as a whole stands perhaps for another consideration as a part; what for one consideration stands as a stage of development stands perhaps for another consideration as a stage of dissolution. A closer entering upon the questions arising here would belong under the discussion of the relation subject—object, or else under the discussion of the concept of relation in the domain of the concepts of value, and must be postponed to later sections. —

Carneades has proved right. We know reality only through the fixed and more or less transparent relations between the objects observation exhibits. The progress of modern science in relation to the old Academics and Sceptics rests on the exactness with which the individual relations are determined, on the increasing number of relations demonstrated, and on the interplay of these relations that is discovered. Thereby the concept of reality has step by step been developed and extended. The Academics did not see the possibilities of development which the concept of relation offered. They could not look with humour on the imperfections of knowledge, because they did not see the significance of a progressive application of the concept of relation. A peculiar form of humour has, on the other hand, become possible in more recent times, in that the great ideal of knowledge is held fast without overvaluation of each step that is taken.³⁹

But to this comes further that behind the Academics' sceptical attitude there lay at bottom always the popular concept of reality, according to which reality stands fixed once for all as an absolute order of things with which our thoughts must agree in order that our knowledge may become true. It is an impossible relation that is here assumed. Reality cannot itself be a term in a relation; that would indeed presuppose that we already knew it. Reality rests, on the contrary, itself on, or makes itself known in, the relations between our observations. Our thought cannot set its one compass-leg on itself and the other on something not yet accessible to it at all. Truth is not unveiling, but bringing forth, and is won only through constant labour.

³⁹Cf. *Den store Humor*. Kap. 6 (Forstaaelse og Humor) og Kap. 9 § 54 (Humor og kritisk Realisme).

d. The Relation Subject—Object

α. Introduction.

The relations hitherto discussed subsist between objects among themselves and express from various sides the fact that human thought works by setting one object in relation to other objects and in that way finds the understanding it can win. The point is at last to lead all objects back to certain fundamental objects which for the time being stand fixed, even if a later age's inquiry may seek to reach further back. As ERNST MACH (*Erhaltung der Arbeit*. 1872) has expressed it: that one stops precisely at certain definite objects can be explained only historically. He adduces a great example (p. 32): "When we nowadays believe that mechanical facts are more intelligible than other facts, . . . it is an illusion. It rests on the fact that the history of mechanics is older and richer than that of physics, and that we have for a longer time stood on a familiar footing with mechanical facts. Who dares assert that electrical and thermal phenomena will not one day stand before us in the same way, when we have come to know their simplest rules and have become familiar with them? . . . A fundamental fact is never more intelligible than another". — That the place of the fundamental facts (or, as I call them, the fundamental objects) in our knowledge can be explained only historically means that it is through the history of science that we can come to understand why at a given time one had to stop precisely at certain definite facts and find full explanation by leading other facts back to them. Mach has in the statement cited foretold what has really happened. In the twentieth century natural science no longer regards inertia, weight and the persistence of energy as fundamental facts. It at any rate sets itself the task of leading them back to other facts. — In philosophy one has often wished to rest content with a certain number of fundamental concepts (categories), and even Kant thought he had here reached the end. But it shows itself that the fundamental concepts knowledge employs in its work are as little the same at all times as natural science's fundamental facts, even though they do exhibit a certain persistent type through the ages.

It is understandable that investigators in the midst of their energetic work do not always have at the same time clearly before consciousness the historically conditioned

character of the limits they reach. Meyerson expresses this thus: science is ontological, only that it puts another ontology in place of that of sound common sense, whether the new ontology appears as atomism, as energetics or as pure mathematics.⁴⁰ — It is, however, not necessary that naive ontology should always characterize scientific work. In the first place, as Meyerson himself shows⁴¹, science itself dissolves the ontology which for the time being seemed necessary. In the second place, the question must naturally emerge, even for the most enthusiastic investigator, with what right he now thinks he has attained “the thing in itself”; that is no foreign question lying outside science itself, as the history of science sufficiently testifies. And in the third place the question must inevitably come forward how, from the ontology one has reached, one can derive retrogressively the world of qualities and successions from which one climbed up to what one thought to have attained as the absolute order of things. This question has the same justification now as it had in antiquity towards Plato and Democritus. —

But besides the objective relations through which objects are illuminated and determined in relation to one another, there constantly asserts itself, though more hidden, a relation between the knowing subject and the objects that are its objects. The ancient Sceptics already recognized that here was a fundamental relation. (See above p. 9). All objective relations are apprehended and thought through by a human subject and hold, for the time being, in relation to it. With a change of this subject, its organization and its conditions, its knowledge too must be able to become another. Here, as with the objective fundamental objects, there is a historical task, namely to demonstrate why the knowing subject meets with precisely certain presuppositions and puts certain questions. This task concerns comparative psychology and the history of science. Here we have only to do with the general point of view: without subject no object.

The converse point of view holds too, however: without object no subject. For we never have a pure subject without objective content and without determination

⁴⁰La science de nos jours est saturée d'ontologie, et les savants, en dépit de ce qu'ils affirment expressément eux-mêmes, font de la métaphysique comme ont fait, de tout temps, leurs devanciers. *De l'Explication dans les Sciences*. II, p. 174.

⁴¹La science, en menant ses explications jusqu'au bout, finit par détruire cette ontologie qui, d'abord, paraissait lui être indispensable. I, p. 50.

and conditioning by this content, just as little as we ever have a pure object, but always (if not with full consciousness) presuppose a possible observer or thinker. The concepts subject and object are correlative, like synthesis and relation, continuity and discontinuity, likeness and difference, and so on. — Every presentation of a world's coming into being, from the Mosaic history of creation to modern cosmogonic attempts, gives a description of how the whole would unfold itself for a spectator with certain senses and certain presuppositions. The knowing subject, the spectator, the thinker, has himself a history determined by the experiences he has had and the way in which he has worked over these experiences. Modern epistemology is clear that through scientific work there can develop a subjectivity with the capacity and the need to see objects in their relations and with an understanding of how it has itself reached the standpoint from which it sets its tasks and demands answers to its questions.⁴² Such a subjectivity may be called the epistemological subject. Heinrich Rickert, who has clearly illuminated this point, aptly calls the epistemological subject a boundary concept in relation to the psychological subject.⁴³ That no doctrine of the categories can be definitive follows precisely from the fact that this boundary concept can and must always be determined anew, because it develops in relation to the objective content present at each time. The world is understood from man's nature and standpoint, but man's nature and standpoint are again understood from the laws of the world and the development of the world. Beyond this interplay one does not get, even if the two points of view may at different times appear with different weight in scientific discussion.

The weight laid on the relation subject—object leads by no means to subjectivism, since, as has been said, there is no pure subject that should be able to bring forth everything out of itself. There is always only an objectively determined S (an S_o), just as there is always only a subjectively determined O (an O_s). That this relation does not come to consciousness without further ado is a consequence of the involuntary trust with which we from the first meet every sensation, every memory, every fantasy, every thought. On this is grounded the thought which the Greeks found it so hard to

⁴²On the reciprocal action between the psychological and the epistemological way of consideration on this point see *Den menneskelige Tanke*, p. 297–303.

⁴³*Grenzen der naturwissenschaftlichen Begriffsbildung*², p. 134–138; 308.

get away from, and which still haunts even clear heads: what I know must after all be something in itself; were it nothing, there would be no knowledge either! — Nor is there at first any reason for doubt about the very forms of thought in which we discuss our problems. And for the time being we do not think that we are ourselves a compass-leg standing at a definite place, from which the other compass-leg can be placed at various places, but always in relation to the first compass-leg. We do not at once discover that we are a term in a relation. Ancient scepticism discovered it. And we shall bring forward some examples of how in more recent times one has more and more come to see the significance of this relation for all our knowledge. What particular part the subjective relation (as we may call it for brevity) plays in the individual case cannot of course be said in advance. It is through analysis of the advancing knowledge of the world that the share which man (the subject) has in this knowledge can gradually become clear. It is some examples from the history of the more recent age's natural science and epistemology that shall now be brought forward.

β. On Copernicanism.⁴⁴

One of the grounds on which Copernicus built his hypothesis was the subjective concept of relation. Sensation cannot, he says, immediately tell us what it is that moves — whether it is the sensed thing, or the sensing subject, or perhaps both with different velocity or in different direction. If now the earth moved, the sensing subjects would move too, and there would be no ground for ascribing motion to the sun. A century before Copernicus a similar train of thought had already been asserted by the profound thinker Nicolaus Cusanus, who drew attention to the fact that wherever a man is thought to be, he will believe himself to be at the centre, no matter whether he is somewhere on the earth or on the sun or on other stars. One has therefore no right to speak of a midpoint in the world, and thus no right either to assert that the earth is this midpoint. And from this it again follows that one has no right to assert that the earth is at rest. When we notice no motion, this may after all come from our not having any absolutely resting point in relation to which it could come forward. This

⁴⁴In this and the next section I build on my presentation of the history of the more recent philosophy.

train of thought is in Cusanus a member of a whole way of seeing which impresses the determination through relation (relativity) of all thoughts, and this way of seeing is again grounded in the fact that all knowledge comes about by the connexion of a plurality of objects, which connexion necessarily brings these objects into relation to one another and to reciprocal determination by this relation. Cusanus has on the whole a remarkable capacity for thinking every object in its definite relations. He thereby stands as a precursor of Copernicus, who showed what new world-picture could be the consequence of this way of seeing, and of Giordano Bruno, who like Cusanus derives the concept of relation from the nature of our knowledge (*del modo nostro de intendere*). Our thought constantly works by determining one object by another, by setting limits and cancelling limits — in virtue of one and the same principle. Thus the one place is determined in relation to the other, and motion in relation to a presupposed fixed point. If I seek such a point on the sun, a motion will look otherwise than when I seek it on the earth. The old world-picture presupposed what was precisely to be proved, that the earth was the one fixed point in relation to which all motion was to be determined in order to be true. Of centre, pole, zenith and nadir we can speak absolutely only when we assume the earth to be at rest. Of particular interest is that for Bruno the relativity of time follows with that of place and motion. Aristotle had (*Physica* IV, 14) taught that the concepts time and motion hang closely together and are reciprocally measured by one another, and for him the measure was given in the regular motions of the heavenly bodies. To this Bruno remarks⁴⁵ that since one and the same motion must look different from the different globes in space, there must be as many different times as there are stars! — Interesting in Bruno is on the whole the way in which the concept of relation leads him to demand new observations and new experiments where one had hitherto rested content with the immediately given as containing its explanation in itself.

In Galileo there enters the turn that a sharp distinction is drawn between the mathematical and the sensuous apprehension. In mathematical knowledge the difference between divine and human thought falls away for Galileo. It is his studies of Plato that are at work when he assumes that in mathematics the difference between

⁴⁵*Acrotismus*, Art. 38 (*Opera latina*. Ed. Fiorentino. I, p. 143–146).

existence itself and our apprehension does not hold: here human knowledge shares in the necessity with which the Deity thinks the truths that bear the connexion of existence; only this difference remains, that what men must laboriously and successively work to understand comes forward for the Deity in immediate intuition, and that the compass of divine knowledge far surpasses that of human knowledge. Only for the mathematical (divine) apprehension is there absolute space and absolute motion. For our sensuous apprehension all motion is relative, since after the overthrow of the ancient, bounded world-picture we have nothing absolutely unmoved in relation to which motion can be established. Our sensuous apprehension is conditioned by our standpoint on the earth: "Think the earth away, and there is no sunrise or sunset, no meridian, no day and no night!" And here on earth it is again our bodily organism that conditions the properties we — apart from the purely mathematical ones (*primi e reali accidenti*) — ascribe to things, thus all the proper sense-qualities (colour, smell, taste and so on). Take the body away, and all sense-qualities fall away too, and then our knowledge will be able to coincide entirely with divine knowledge.

The sharp opposition between mathematical (divine) knowledge, for which there is absolute space and absolute motion, and sensuous (human) knowledge is the more remarkable to find in Galileo, since his great deed was precisely the methodical co-operation of deduction and induction, of mathematics and experiment. But this opposition gained great influence in the following centuries, especially as it was supported by Newton's great authority.

On account of the proposition of the relativity of motion, the celestial phenomena could purely formally (mathematically) be explained just as well by the sun going round the earth as by the earth going round the sun. And Tycho Brahe had set up an intermediate form between the two hypotheses, according to which the earth stood fixed while the sun and the planets went round it. Gassendi instituted a comparison between the three hypotheses thus set up and sought to show that purely scientifically one could not choose between them. But then Newton showed in his famous work that the laws Kepler had discovered for planetary motion could be derived if one extended the law of gravitation to hold for the heavenly bodies, and then it must necessarily be the earth and the planets that moved about the sun. In virtue of the

law of falling motion, then, only one hypothesis was at last possible. In an analogous way Leibniz at about the same time led beyond the scepticism which the relativity of motion apparently motivated, in that he maintained that what was decisive for knowledge was not motion itself but the laws of motion, and here he then set up — in criticism of Descartes' doctrine of the persistence of motion — the proposition of the persistence of force. Here then was a reality that did not conform to the observer's standpoint in space. The two great investigators led, each by his own path, beyond the apparent subjectivism and took the decisive standpoint in the laws of motion.

Nevertheless Newton maintained (here surely under the influence of the Platonist Henry More), like Galileo, a sharp opposition between sensuous space, to which the popular apprehension holds, but which makes no absolute determination of place possible and exhibits only relative motion, and absolute space, which stands in no external relation whatever, and which permits absolute determination of place and absolute motion. It is in scientific orientation in the world (in *rebus philosophicis*) that we make use of the absolute places (*loca primaria*), and it is then mathematical space that lies at the basis. In practical life (in *rebus humanis*), on the other hand, we content ourselves with sensuous space, which makes no absolute determination possible. This conception, which hangs together with Newton's peculiar metaphysics, became for the time being the dominant one within natural science. It was a formidable barrier that was by the great naturalist's doctrine on this point set against the continued application of the concept of relation. Since one no longer, as Copernicus still did, assumed the sphere of the fixed stars ("the eighth sphere") to stand fixed, so that it — as *communis universorum locus* (to use Copernicus' expression) — could serve as basis (*point de repère*) for absolute determination of place, it was really a purely mystical reference to appeal to absolute space, which was not accessible to any human experience. — At this point of the discussion two philosophical thinkers now intervene in a peculiar way.

γ. Philosophical Standpoints.

It was a new Platonism that triumphed with Galileo and Newton, in peculiar opposition to the way in which these great investigators in their natural science applied the

reciprocal action between thinking and observation. But an opposition now arises which may be compared with that raised in antiquity by Carneades and Aenesidemus against the Platonic doctrine of Ideas. It was Berkeley and Kant who raised this opposition. They both criticize, each in his own way, absolute space, and maintain that if one leaves the knowing subject out of consideration, all truths and all riddles fall away at the same time.

Berkeley's principal thought was that we think and must think in examples. General concepts, propositions and laws we know only from the special cases in which they display themselves, and it is unjustifiable to ascribe to them and to the relations they express existence as a world of their own behind the world of qualities and differences of quality appearing to observation. The "primary and real" properties, to which Galileo ascribed absolute existence, we nonetheless always apprehend only under definite conditions: at some distance or other, macroscopically or microscopically, and so on. Extension in space must share the fate of all other properties. Every motion must have a definite direction and a definite velocity. There is neither extension nor motion in complete generality. When the question is raised about the right place, the right magnitude, direction or velocity, such a question can be decided only with regard to the apprehending and thinking subject. If I, says Berkeley, being the childlike believer he was, had been present at the creation, I should have seen things come into being in the order the Bible relates. But in daily speech we disregard the subject, although it must always be understood. — Here the opposition to Newton appears clearly; Newton thought that in strict science we disregard the subject. Berkeley maintains that in science (in *rebus philosophicis*, to use Newton's expression) a subject must be understood just as much as in practical life (in *rebus humanis*).

Kant has, as already mentioned, seen that relation is a fundamental concept holding for all knowledge, only that he, with decided inconsistency, reserved a relationless "Ding an sich" to explain that there exists for us something we have not ourselves brought forth.⁴⁶ Among the relations it is for him, as for Berkeley, the relation

⁴⁶When "Ding an sich" is to contain the ground of matter (and form) in our knowledge, it nonetheless stands in relation to knowledge, notwithstanding that it was to be relationless. Or our experience stands related to it, but it not to our experience. It is the same one-sided relation that Thomas Aquinas tried to carry through in theology. See above p. 47 Note.

subject—object that occupies the interest, and he maintains the fundamental position of the subject in this relation. Even in pure mathematics and exact empirical science a knowing subject is presupposed. Only for such a subject is there experience in the strict sense of the word, an experience in which mathematical concepts find their application and have their real significance.

Kant began as an eager Newtonian and for the time being maintained, like his master, absolute space as an objective presupposition for strict empirical science as well as for pure geometry (which for him had attained its definitive foundation with Euclid). Even in “*Kritik der reinen Vernunft*” he maintains an absolute space, only that the Newtonian space has so to speak been “turned inward” and become a human form of intuition instead of a divine form of intuition, a *sensorium hominis* instead of, as for Newton, a *sensorium dei*.⁴⁷ Of particular interest, however, is that he emphasizes that the concept of space (and the same holds of the concept of time) is not merely a general concept, but what one might call a typical individual concept: every definite space is a part of space, not merely an example of the concept space. And it must also be noted that when he calls space a form of intuition, he does not mean by intuition a result finished once for all, but a function, an intuiting, which leads to the forming of intuitive images or the construction of schemata. On the whole Kant means by “forms” ways in which the connecting function, the synthesis in which all consciousness and all knowledge consists, takes place in the individual cases. We construct space by adding point to point, piece to piece according to a definite rule. It is in the identity of such a rule that the unity of consciousness expresses itself in the individual case.

By a closer carrying through of this train of thought Kant later came to deny the subjective reality of absolute space, just as he had already in “*Kritik der reinen Vernunft*” denied its reality as existing in itself. Already in the first edition of his principal work he had stated that, apart from all things that fill and limit, absolute space is nothing other than “*die blosse Möglichkeit äusserer Erscheinungen*”. Some years later (in “*Metaphysische Anfangsgründe der Naturwissenschaft*”) he goes a step further, relying on the relativity of every place and every motion. In absolute space it should indeed be possible to disregard all relations. But when do we have done with

⁴⁷Cf. *Om Kontinuiteten i Kants filosofiske Udvikling*, p. 31.

setting all points in relation to one another, with drawing all lines and so on? Where is the point or the limit which, itself relationless, determines all points and limits? Kant therefore declares that pure, absolute space “für alle mögliche Erfahrung nichts ist”: “Der absolute Raum ist an sich nichts und gar kein Objekt, sondern bedeutet nur einen jeden relativen Raum, den ich mir ausser dem gegebenen jederzeit denken kann.” (Anf. Skr. 1786, p. 16 cf. p. 3). Kant here in fact espouses Berkeley’s doctrine that we think in examples. Absolute space means merely the negative fact that no given space is conclusive. New spaces can always be imagined. It is the very thought of relation, of relation-setting, that leads Kant away from his absolute form of intuition. Space becomes an “Idea” in Kant’s sense, a boundary concept which rejects conclusion and maintains the possibility of continued formation of series. The concept of space acquires only regulative significance, like the “Ideas” set up by Kant himself, and at bottom it belongs under “the cosmological Idea” (the concept world).

Kant has himself compared his philosophy with Copernicanism. And in a way he and Berkeley might have been discussed in the section that dealt with the thoughts developing “on Copernicanism”. But the two philosophers were nonetheless mostly occupied with combating the dogmatism which, with supposed support in mathematical science, had maintained itself in spite of the victory of Copernicanism. And as regards Kant in particular, his principal endeavour was at one and the same time to maintain the great significance of the Newtonian concept of science and to lead the absolute forms Newton had still let stand back to being forms of the working thought.

Although Kant constantly set time parallel with space, he does not for time draw a similar consequence as with respect to space. And yet it would — particularly with his remarkable doctrine of the “schematisms”, which emphasizes the activity in the intuition of time — have lain near him to see that we have as little done with time as with space, and that absolute time can mean only that new relations of time are always possible. —

I need not here enter upon a criticism of Kant. Such a criticism would particularly aim at showing that the very concept of experience in Kant’s strict sense indicated above is an “Idea”, like space and time, and that Kant can assume a “Ding an sich” without relations only by a strange inconsistency.

δ. The New Theory of Relativity

Despite the great expansion of the horizon that Copernicanism gave impetus to, natural science had held fast to the concepts of absolute place and absolute motion, which had their justification only within the old worldview with its bounded and fixed frame. Galileo's and Newton's absolute space replaced the space bounded by the eighth sphere. But there came a time when the concept of relation here too made its consequences felt, and not only with respect to the relations among objects, but also with respect to the relation between objects and the knowing subject.

It was the first proposition of kinematics, and thereby of natural science, that was first examined from the standpoint of the concept of relation. NEUMANN (*Ueber die Prinzipien der Galilei-Newton'schen Theorie*, 1870) drew attention to the fact that the law of inertia, according to which the velocity and direction of a body change only through external influence, necessarily presupposes in each individual case the specification of the particular place in relation to which velocity and direction are preserved when no external influence is exerted. Neumann could find no place that could lay claim to being taken as the starting point for an absolute coordinate system, but held that one must hypothetically assume a center in order for the concept of absolute motion to retain its meaning. MACH, who at roughly the same time had arrived by his own path at the insight that the law of inertia rested on the presupposition of a determinate coordinate system, did not find Neumann's solution satisfying. He demanded that the concept of relation be carried through. The dogmatic conception of the law of inertia, in his view, only falls away when one becomes clear that in itself any body in the world can be used as a starting point, and that in practice in each individual case we use one body or another as the presupposed fixed starting point for the coordinates in relation to which the velocity and direction of a motion are determined. (*Die Erhaltung der Arbeit*, 1872). This is the path that has in fact always been taken, only within the old worldview, which rested precisely on the presupposition of an absolute center and an absolutely fixed boundary-sphere of the world, no choice was required. Mach lays great weight throughout on the historical presuppositions on which the research of each period rests, but then shows how these presuppositions vary from age to age, and that none of them can lay claim to absolute

necessity and intelligibility. What matters here always comes down to what French physicists call points de repère.⁴⁸

An even more radical application of the concept of relation appeared at the beginning of the twentieth century. What is of interest in this connection is that it has proved necessary to take special account of the apprehending subject and its standpoint. Copernicus had already pressed this point as regards the apprehension of the motions of celestial bodies, and Neumann and Mach as regards the velocity and direction of motion. EINSTEIN⁴⁹ has now sought to show that what holds for space and motion holds also for time. That this was not discovered earlier is, according to Einstein, to be explained by the fact that attention was paid only to velocities that were very small in relation to the velocity of light. The final temporal measure becomes, for modern physics, the velocity of light, whereas for Aristotle, and even for the Copernicans, it was drawn from the motions of celestial bodies. (Bruno's doctrine of the relativity of the concept of time rests on this measure. See above.) On the basis of the electromagnetic theory of physical phenomena, for which the mechanical conception of nature stands as only a special case, Einstein seeks to show that there can be demonstrated circumstances under which two observers, of whom one is attached to fixed and the other to moving axes, but both at rest relative to their own axes, will not arrive at the same apprehension of time: what for one observer stands as a relation of simultaneity between two events, will for the other stand as a relation of succession. A temporal specification acquires meaning only when it is stated in relation to which coordinate system it holds, and in particular whether that system is itself relatively at rest or in motion.

If one were to hold that an illusion must be present in one or the other observer, the answer from the Einsteinian standpoint is that each observer has his own system of measurement, which is just as legitimate as another's, though it is of course perfectly possible for one observer to attempt to place himself at another's standpoint and reason from it.⁵⁰

⁴⁸Cf. e.g. PAINLEVÉ (*Mécanique*, p. 399) [in the collective work "De la Méthode dans les Sciences", 1909]: "... les mouvements absolus, c'est-à-dire au fond les mouvements convenablement repérés." [i.e., absolute motions are, at bottom, motions suitably referred to reference points.]

⁴⁹*Über die spezielle und die allgemeine Relativitätstheorie*, (1916)

⁵⁰LANGEVIN in "Bulletin de la Société française de Philosophie" 11 October 1911. — It must presumably

The facts and inferences on which the Einsteinian theory rests are still a matter of dispute, and views stand sharply opposed to one another. In such a dispute the philosopher as such cannot intervene, and philosophy has time to wait. Whether physics can really demonstrate situations in which what stands as simultaneity for one observer stands for another observer as successive, physicists must determine among themselves. If this must be answered with yes, we stand before the significant result that natural science has once again, as with the Copernican hypothesis, by its own path come to see the necessity of not merely holding to the content of observations, but of taking account of the observer's presuppositions. In an interesting discussion in the "Aristotelian Society" (1919), WHITEHEAD impressed upon us the significance of what he aptly called "the percipient event"⁵¹, and NICHOLSON declared that among the data that are of decisive significance (ultimate character) there must also be counted "the essential framework of our modes of perception."

Against the Einsteinian hypothesis, as against earlier demonstrations of the epistemological significance of the concept of relation,⁵² it has been urged that there can only be one truth in each individual case, and that it would contradict the principle of non-contradiction if two observers, of whom one declares simultaneous what the other declares successive, were both to be right⁵³. But there is no reason to fear for the unity of truth, even if Einstein should be right. The real truth lies in this case in the necessity, physically demonstrated, that different observers under certain determinate conditions arrive at different apprehensions of time. Each individual observer apprehends time as he must under his conditions. But if an observer is also a physicist, he will (assuming the correctness of Einstein's theory) be able to explain to us why another observer arrives at a different apprehension than he himself does. (Cf. Newton's and Leibniz's emphasis on the lawfulness of motion as the expression of its objectivity. Above, p. 66.) Naturally, the physical explanation of the different observers' different apprehensions

be assumed that the observer who can place himself at another observer's standpoint knows sufficient physics.

⁵¹See also his *Principles of Natural Knowledge* (1919), p. 68. — Whitehead's conception differs from Einstein's, however, in that "the percipient event" for him is only one factor alongside many others, and not, as in Einstein, of essential significance for all measurement of time and space. Cf. on this point HALDANE: *The Reign of Relativity* (1921), p. 69 f.; 107 ff.

⁵²Cf. *Den menneskelige Tanke* [The Human Mind], p. 304.

⁵³KROMAN in *Tidsskrift for Fysik*, XIV, p. 15.

of time itself rests on certain presuppositions, presupposes a coordinate system from which the coordinate systems of the various observers are determined, and here there is the possibility of a new discussion. When this discussion comes to an end, it is because no standpoints can be demonstrated that are still more fundamental. Kant long since maintained that all necessity is hypothetical. But it is necessity nonetheless — or rather: precisely because of this it is necessity. A necessity without presuppositions we do not know.

If the new theory of relativity wins continued confirmation, what comes to hand will not merely be a result of physical research that is interesting in and of itself. This theory also contains, as Einstein himself has declared, a spur to continued inquiry. If it is the case that there is the possibility of different systems of relations in the apprehension of an object, this constitutes precisely an invitation to discover all the particular relations that lie at the basis in each individual case, and inquiry is thus led ever further rather than closing off in dogmatic fashion, as if the presuppositions one had hitherto worked with were the only possible ones. There are, from an epistemological point of view, two interests that attach to the concept of relation. One is that light is cast on the nature and conditions of our knowledge; the other is that new tasks are indicated, a plus ultra appears, in that every thought points beyond itself. —

Campanella said in his apology for Galileo that if Galileo were right, one would have to philosophize in a new way. But if Einstein is right, one need not therefore philosophize in a new way.⁵⁴ Even from a purely philosophical point of view, as the foregoing presentation has shown, attention was long since drawn to the significance of the concept of relation for knowledge and worldview. But if Einstein is right, there now lies before us a new, unexpected, but significant experience in that respect. —

Through the necessity of taking account of the knowing subject, and through the possibility of a conflict between the each-for-itself necessary apprehensions of different knowing subjects, the newer natural science — in Einstein as in Copernicus

⁵⁴EINSTEIN himself agrees with this. In his preface to LUCIEN FABRE's book "Les théories d'Einstein" (1921) he says: "The theory of relativity neither can nor will provide a world-system, but only indicate a limitation to which the laws of nature are subject." That it is not merely a matter of a limit, we have seen above.

— offers an analogy to what asserts itself powerfully in the domain of the human sciences, as I shall now attempt to show.

4. The Concept of Relation in Ethics

a. Historical Research and the Concept of Relation.

Aristotle taught, as already mentioned, that only the general, not the individual as such, could be an object of science. This hangs together with the logical-schematic character of his philosophy. Science was for him, as for Plato, subordination of the particular and individual under general concepts. But now for Aristotle everything real was at the same time particular and individual; it is on this point that he comes into opposition to Plato. The difficulty arising hereby, and which has been brilliantly illuminated by ZELLER (*Die Philosophie der Griechen*, II, 2), can be overcome only through a closer discussion of what one understands, and scientifically must understand, by reality; and such a discussion was, as we have seen, introduced with great energy by the Academics and Sceptics of antiquity.⁵⁵

Already here the thought came forward that the criterion of reality can consist only in the fixed connexion between observations. Just as the formal sciences spring from a striving to preserve agreement with oneself, so the real sciences spring from a striving to find agreement between the objects given to observation. And the criterion of reality lying in this must hold just as well for the individual, that which perhaps occurs only a single time, as for that which can repeat itself again and again because it holds for whole series of individual objects.

An object is known as real, partly through its connexion with other objects concerning which no doubt has been awakened, partly through the connexion that can be found between the various forms under which the object itself appears at different times, and between the various properties with which it appears at a given time.

⁵⁵It is another matter how the difficulty could be cleared up within the Aristotelian philosophy itself. HANS V. ARNIM (*Die europäische Philosophie des Altertums*. Kultur der Gegenw. I, 6, p. 173 f.) finds the explanation in the fact that both species and individuals are in Aristotle regarded as “beings”. But thereby a new question is merely called forth, namely how and why the one kind of “beings” can be an object of science and the other not.

Historical research now aims precisely at both kinds of connexion. The events, personalities or institutions that are its objects it investigates in relation to the circumstances under which they are present, and each single one of those objects it investigates with regard to its inner connexion. In both respects a beginning is made with criticism of reports, but this criticism itself lays the named criterion of reality at the basis. Thereafter it becomes the task to find the bond connecting the object with other objects, and the bond that ties together the object's various sides and stages.

The object of science is neither the general nor the individual in and for itself alone, but reality, which can be known only through the bonds uniting objects among themselves as well as the elements within the single object.⁵⁶ What cannot be illuminated through application of the concept of relation cannot be understood.

Philosophers of history have in the most recent time brought forward three concepts as particularly characteristic of historical inquiry⁵⁷, and these three concepts express precisely relations between an object, its conditions or its effects.

Historical centre is the definite event, personality or institution whose origin, subsistence and development the historian wishes to investigate. About it one gathers everything that can contribute to its illumination, but takes no account of what neither furthers nor hinders it. Correlative to this concept is the concept historical threshold, which denotes the limit of what can be included in the individual investigation. Historical greatness denotes that character of an event, a personality or an institution whereby there are expressed in them forces working in the direction of the solution of new, significant social tasks.

Now there stands in history as in nature a struggle for existence. The historian stands before events and personalities in the human world as the biologist stands before plants and animals in their struggle for existence. The point in both cases is to understand the contending parties each for itself, as regards their nature and their conditions, and thereby to understand how a struggle arises between them. The biologist as such does not "side with" any of the organic forms, even if his intellectual interest has made a single one of these forms the centre of his investigations, or even if he can feel a certain sympathy for certain forms of life, as when it gladdened Darwin

⁵⁶Cf. *Totalitet som Kategori*, p. 59–66.

⁵⁷Anf. Skr. p. 71–72.

when he thought he could find functions in plants that might point to a higher stage of life than one was accustomed to ascribe to them. For the biologist the point is merely to understand the necessity of the struggle under the given conditions. Nor does the historian as such take sides, but seeks to find what conditions assert themselves in each of the contending parties, in order to understand how interests, thoughts and feelings have had to unfold themselves in a definite degree and in a definite direction in each of them, until the collision became unavoidable. Every nation, says William James in one of his letters, has its ideals, which are a dead secret to other nations, and must under the influence of these ideals develop in its own peculiar way. If this is so, there is required of historians an understanding both of how these ideals themselves have developed and of how they have determined the national development. Thereby it again becomes possible to understand the tension and the strife between nations. It has been said of Leopold Ranke that when in his historical inquiry he avoided one-sided partisanship, it was not because he was neutral, in the sense of indifference towards all contending parties, but because he was universal and could enter into and follow the life stirring in each of them. This historical universality is akin to the artistic universality that enabled Shakespeare to let each of his figures unfold according to their own inner laws. His impartiality towards persons and fates sprang not from indifference and remoteness, but from being seized by life as it can stir in different men and lead them to collisions that perhaps become tragic.

The historian (and the poet) stands here in principle as the physicist stands when he establishes certain relations under which what for one observer stands as simultaneous stands for another observer as successive.

b. Ethics and the Psychological Relations.

Historical valuation is not properly an ethical valuation. It concerns the significance of an event, a personality or an institution for the continued actual development, but does not pass its judgments from an ethical ideal. With great fineness and energy HEINRICH RICKERT has in his work "Grenzen der naturwissenschaftlichen Begriffsbildung" demonstrated this difference between history and ethics. And from the historical side it has been energetically stated that historical inquiry does not penetrate to the

place where actions and endeavours have their source: “Das tiefinnerste Geheimnis einer Seele zu finden, damit deren sittlichen Wert, das will sagen, den ganzen Wert der Person richtend zu bestimmen, hat die historische Forschung keine Methode und keine Kompetenz”. (DROYSEN: *Geschichte Alexanders der Grossen*, p. 259).

If nevertheless the ethical standpoint of value involuntarily forces itself forward in the historian when he tries to give his presentation its final rounding-off, he will stand before the difficulties of principle presenting themselves in every attempt to ground ethical judgments, thus in every ethics.

At this point a new class of categories appears, which does not coincide with any of the classes — the fundamental, the formal, the real — that we have hitherto discussed: the concepts of value or the ideal categories. The concept of relation asserts itself here at once, in that value can be determined only in relation to value. All willing is a preferring, and every feeling has its degree and its quality determined by its relation to other feelings; therefore in every valuation everything comes down to what stands beforehand or simultaneously as valuable. At last all valuation in the individual cases will rest on the relation to a fundamental value. Just as a place can be determined only by its relation to other places, a time only through its relation to other times, a reality only through its relation to other realities, so every determination of value presupposes that certain other values stand fixed beforehand. The fundamental value corresponds to the globe or the coordinate system from which we orient ourselves in space, — to the event in relation to which we date other events, — to the reality which in the individual case decides what is to be reality for us. And if one investigates more closely the various possible fundamental values asserting themselves in human valuations, one will find that they all presuppose some whole or other whose subsistence and development comes to stand as end and thereby determines the valuation. Such a whole may be the valuing personality itself, or a society (family, estate, state, church), or a sum of endeavours (art, science and so on). A valuation will at last be determined by an action’s relation to the conditions of such a whole’s subsistence and development. Only when a whole given in experience is presupposed, whose conditions of existence and development can be discussed, is a rational valuation, and thus a rational ethics,

possible.⁵⁸ Ethical idealism often works against itself by seeking beyond all relations. In the several places where Plato discusses “the Idea of the Good”, it is expressly raised (like other Ideas) above the world of relations. He does indeed maintain (especially in “Gorgias” and “Philebus”) a connexion between the valuable and definite relations of measure, namely proportionality (“geometrical likeness”) or harmony between fullness and limitation. But such relations of measure hover in the air when they are not derived from the conditions of life of a certain definite whole, and here Plato oscillates between an individual and a social totality.

The concept norm concerns the means by which a presupposed whole can subsist and develop. The presupposed whole then stands more or less consciously as end, in that its value denotes a fundamental value. The concepts value, end and norm lie in the same series, and in a rational ethics they must lie in this definite order. No ethics can withdraw itself from the connexion between these three concepts. Often one conceals the first term in the series and holds either to the concept end or to the concept norm, as though they were primary. We meet here again that lack of consciousness of where the one compass-leg stands which so often leads to a misjudging of the significance of the concept of relation. One then regards the position of the other compass-leg as absolutely given, self-evident or “objective”. Not least does this appear in moralists who think they have secured themselves an “objective” standpoint by proceeding from “society” (without even saying what society they mean). Or it is, as in Kant, the concept norm that one lays at the basis. Kant, who in his epistemology is clear that every necessity is hypothetical, maintains an ethical necessity (a categorical imperative) which is not again to rest on any presupposition, and in various forms later moralists have followed him in this. When an ethical system is criticized, the criticism will — apart from an investigation of the system’s formal consistency — essentially aim at showing that the system presupposes a fundamental value which is not itself drawn into the investigations. Criticism will seek the hidden compass-leg. Even he who proclaims a “revaluation of all values” must presuppose some value or other in relation to which he undertakes the revaluation.

The fundamental value will at last be determined by the impulse to self-preservation

⁵⁸On the connexion between the concept of totality and the problem of valuation see: *Den menneskelige Tanke*, p. 237–241; 346–369. *Totalitet som Kategori*. Kap. 5.

in the individual (in so far as it can be thought isolated) or in the greater or smaller society with which the individual feels himself in solidarity, so that this society's weal and woe, and thereby its ends, become the individual's too and determine his ethical tasks. There may, however, also be expressed a striving to make the single moment or a single side of the individual's life a whole in and for itself. From the single moment to the human race in present and future there is a series of possible foundations for ethical systems, and these foundations are, as the history of ethical ideas shows, in part made use of.

The comparative moralist now has (as in their domains the physicist and the historian have) the task of understanding the conceptions which the various foundations (fundamental values) make possible, and which may come into decisive conflict with one another. Thereby is understood the fact that the same action, the same personality, the same institution can be the object of conflicting ethical judgments. On account of the lawful connexion which can be demonstrated by psychological and historical means in the spiritual world, we seek an explanation of why others judge otherwise than we do, and thus have other values, ends and norms than we.

And it is not merely the comparative moralist who has use for such understanding. For it is after all a concern of every ethical standpoint to get men to know and acknowledge the values, ends and norms which it itself lays at the basis. And a beginning must hereby be made by taking men where they actually stand, entering into their presuppositions in order to be able to lead them to new presuppositions. Socratic irony is, seen from an essential side⁵⁹, a means of getting men to open themselves, to reveal themselves, in order that it may thereby become clear how help, education and co-operation can become possible. It is the analytical method which alone can lead from one standpoint to another, when this is possible at all. It is of course easier to begin with the construction of the "right" ethics and then leave it to men how they can come to terms with it. But every value and every valuation has in virtue of the concept of relation its definite psychological presuppositions, and the point is first and foremost to discover how far these are present, so that a common valuation can become possible. Only a common fundamental value makes common

⁵⁹Cf. *Den store Humor*, p. 63.

valuation possible.

Misjudgment of the relation between what is a moralist's fundamental value and what he is from this foundation led to teach has often led to unjust judgments about ethical thinkers. Thus it has been thought that the weight which philosophers like Adam Smith, Helvetius and Bentham have laid on the acquisitive impulse, on individual interest and on the impulse to happiness, must be due exclusively to a preoccupation with individual or egoistic interests, so that their own fundamental value must consist in such interest. A closer investigation shows, on the contrary (as I have shown in the relevant section of "*Den nyere Filosofis Historie*"), that the named thinkers all proceed from universal sympathy and from conscious regard for the welfare of human societies. Their conception is that society will thrive best by each individual building on his experiences of what serves him best, and that no authority standing outside him can know this better than he himself, when his eyes are opened. Through free labour and through the reciprocal action of independently striving individuals, the fundamental value on which they build will for the named thinkers best come into its own.

The individual moralist must be conscious that the fundamental value which bears his system can itself be brought forward for testing. If the system has a vulnerable point, it will at last lie here, even if the system is otherwise developed with ever so great consistency and supported by ever so many experiences. And he must, when he is sufficiently critical, allow Henry Sidgwick to be right that it — at any rate at certain stages of development — is good that in a society different, even conflicting, ethical conceptions should be asserted, even though he can of course follow only his own.⁶⁰ In the ethical domain too, truth can work its way forward only through strife. Psychology and history have then to see to it that behind the burning strife there can be an underground understanding.

The whole conception of the position of ethics which is developed here and in what follows in connexion with the concept of relation, I have maintained from 1887 onwards in my various writings, and it has constantly won new firmness with me, not least on account of the criticism to which it has been subjected from various quarters.

⁶⁰*Methods of Ethics*,² p. 450.

c. Ethics and the Historical Relations.

The working out and carrying through of ethical norms — thus the determination of the subordinate ends which the principal end leads one to set, and in which the derived values that the fundamental value gives rise to find their expression — is possible only in reciprocal action with the given historical conditions. Neither the valuable qualities of character (“virtues”) nor the social arrangements which a fundamental value requires for its full realization can be constructed in advance; they cannot be “introduced”, as in old days one thought one could “introduce” a religion or a constitution. In any case the historical conditions will at different times and in different places give both individual qualities of character and social arrangements essential peculiarities which cannot be derived solely and alone from the general norms to which the fundamental value gives rise. Even a brief survey of the history of the virtues, such as the one I have attempted to give in my *Ethics* (X, 4), shows this. As regards the social arrangements, it is seen from the history of marriage (Etik XV) and of the social question (Etik XXV). The old leather bottles cannot at once be replaced with new ones. Besides, ethical norms always have a general character, and the specialization can be grounded only on the historical conditions under which they are to be carried through.

Abstractions and vague dreams are not enough. The point is for the time being that the direction for the development of character and the ordering of society should be determined by the fundamental value. A complete programme only experience can lead one to work out. It goes with ethical norms as it goes with the concept of cause in epistemology, which likewise does not in itself show any solution of special questions, but merely motivates an asking and a seeking which acquires a provisional character at various stages of science’s historical development.

There lies here the possibility of tragic conflicts, in that those who are animated by a fundamental value may lack an eye for the historical conditions, while those who stand firmly on the ground of historical reality may lack an eye for values that demand realization. In the present day there stands more than ever a struggle between social ideals and social realities. Bertrand Russell, for whom the social ideal is given in communism, finds that the price which by the application of Bolshevik methods is to be paid by the human race in order to reach communism is far too high and will

even cancel the value which the carrying through of communism would in itself have. A principal cause of this he finds in the fact that in Bolshevism hatred of the old plays a greater part than hope of the new; these psychological moments too are historically conditioned.⁶¹ The counterpart to this is found more than sufficiently represented in Europe as hatred of the new and hope of being able in spite of everything to preserve the old. Of both opposites it holds that they have learned nothing and forgotten nothing. When he who is filled with an ideal also possesses intellectual honesty, he will, for the ideal's own sake, investigate all the conditions under which it is to work, and all the ways in which this working must intervene in human affairs.

The matter now stands thus, that the fundamental values and the norms themselves come into being only under quite definite historical conditions. Ideals which should have arisen and been formed without the influence of historical realities have never existed. Plato's ethical idealism did not form itself in his consciousness without the influence of the past of Greek intellectual life, and his ethics has, in spite of all opposition to his people's custom and to his contemporaries' notions, a decidedly Greek stamp. Christianity has in its original form a decidedly Oriental stamp, and it points back to Jewish tradition and to Persian notions. Kant's ethics has a long and significant history of development. If I am here also to speak of myself, I have expressly emphasized that the presuppositions on which I have built my presentation of ethics are due to historical development and cannot be said to have been present at all times. When WESTERMARCK⁶² says that if the word "ethics" is to be used as the name of a science, its task can only be to study the moral consciousness as a fact, I do not feel myself hit by this remark, which is among other things also directed against me; for I have always assumed a historical-psychological development of the moral consciousness and maintained that every ethics will correspond to a certain form and a certain stage of this process of development. (See Etik III, 13; 21). This does not deprive ethical ideas of their significance or their validity. One can after all steer by the stars, although they have a history. And besides, the ethical voyage takes place on earthly seas and is therefore conditioned not merely by the stars, but by wind and wave.

⁶¹BERTRAND RUSSELL: *The Practice and Theory of Bolshevism*. (1920), p. 146–178.

⁶²*Origin and Development of Moral Ideas*. I, p. 18.

The task must always be a development of what is historically given in the direction of the ideal norms. But there must at the same time take place an orientation with regard to whether these norms are indeed rightly set up, and whether they are indeed really the expression of experiences of value. If for one reason or another the fundamental value itself is shaken, it becomes a psychological-historical question whether transition to another fundamental value is possible. Here too there is the possibility of serious crises and tragic catastrophes.

d. Ethics and the Individual Relations.

Besides the psychological relations, which concern the acknowledgment of a fundamental value and of the norms following from it, and the historical relations, which concern the possibility of carrying these norms through under given conditions, there comes forward in a deeper ethical discussion yet a third kind of relations, which concern the individual's possibility of satisfying the norms. If ethical demands are not to hover in the air, they must be set in such a way that the individual's capacity and need are taken into consideration. Only then will a striving in the direction of what is demanded be able to be released. There cannot therefore be mechanically demanded the same of everyone. Of him who is capable of great or self-sacrificing activity there is demanded something other and more than of him who under inner discord or by reason of powerlessness of mind perhaps just manages to keep his head above water. Only then is the same ethical work in reality demanded of all; for ethical work is the expression of the energy with which inner and outer resistance is overcome, and if this resistance is different for different individuals, the work actually performed can be determined only by the relation to this different resistance. What is demanded of the individual may then lie now above, now below the average demands set by a general law. Everyone is, as Schiller's Wilhelm Tell says, taxed according to ability.

In a treatise on the laws of relation in ethics (*Etiske Undersøgelser*, 1891) I have in detail grounded that individualization of ethical norms which is as necessary as it is a difficult consequence of the concept of relation. It rests on the principle that the law exists for man's sake, not man for the law's sake. The work the individual must do in order to satisfy ethical norms, the virtues and the duties demanded of him, must

themselves serve as means for his personal development, and this presupposes that his capacity and need become co-determining for the demands which according to the norm are to be made of him in particular.

Most ethical theories have misjudged the significance of this individual relation. Nevertheless it has again and again asserted itself. In theology it lies at the basis of the concept grace, in jurisprudence of the concept pardon. For these concepts are unethical when regard for the individual conditions does not lie at the basis of them. In philosophy the analogous concept equity has been set up, which stands in opposition to abstract justice, but in reality, if it is an ethical concept at all, must be said to express the highest justice, since it takes the individual moments into consideration. In the most recent time's individualization of the execution of punishment this consideration also comes forward; what a transgressor of the law is sentenced to is precisely the definite execution of punishment actually applied to him.

From the significance of the individual relations there follows an important consequence for social ethics. It must be a necessary task to strive to provide all members of society with such possibilities of development that the choice of work can become free. So long as this is not attained, the ethical demand upon society has not been satisfied. On this point the social struggle will more and more come to be joined.

e. Kant's Ethics and the Concept of Relation.

Kant, who himself, as we have seen above, saw in the concept of relation a fundamental law for all thinking, would nonetheless not acknowledge it in ethics. The ethical law shall according to him in no respect or degree have proceeded from experience; psychological and historical relation shall thus not hold for it. And it shall demand the same of all, whatever individual presuppositions they may have. But for that reason the ethical law stands, according to Kant, also as an inexplicable marvel before the working, thinking and striving human race, which is constantly placed before definite conditions. Kant states this quite openly. "How pure reason can be practical, human reason is quite incapable of explaining, and all trouble, all labour to seek an explanation of it is wasted". (*Grundlegung*³, p. 125). "How a law can in and for itself and immediately be a determining ground for the will, that is for human reason

an insoluble problem". (*Kritik der praktischen Vernunft*, p. 128). Kant does indeed concede that the law can work in us only by calling forth a feeling of the law's own sublimity; but he then declares this feeling to be psychologically inexplicable.

But from this relationlessness in psychological, historical and individual respects it again follows that we can at bottom pass no properly ethical judgments. When we have no access to understanding the relation between norm, motive and action, we can never have the right to call any human action before us good or evil. This consequence Kant already drew in the first edition of "*Kritik der reinen Vernunft*" (p. 551), though only in a note below the text: "Die eigentliche Moralität der Handlungen bleibt uns daher, selbst die unseres eigenen Verhaltens, gänzlich verborgen".⁶³ This is a correct consequence when the concept of relation is not to be applicable in ethics. The matter stands otherwise when ethical norms are conceived as the expression of endeavours which are precisely formed into more perfect forms through the necessary relations asserting themselves in all thinking and acting. Precisely the concept of relation opens great possibilities for us here. It demands that we set the one compass-leg at a definite place in the real world (in a psychological fact, in something historically given, in an individual possibility) and then try how far we can carry the other compass-leg out. If we cannot at once set the other compass-leg where we wish, we can perhaps gradually move it further out. Perhaps even the first compass-leg can be moved. The point in ethics as in physics is definite points de repère.

Just as space and time are "Ideas" (in Kant's sense of the word), and just as experience (reality) is an "Idea", so too the ethical good is an "Idea", which acquires real significance not by excluding all relations, but by using them to set the special tasks and to judge the attempts at their solution.

Kant and Droysen are nonetheless not entirely wrong in their declarations that the proper morality of actions cannot be discovered or grounded. For we can never be sure that we have taken all relations into consideration. It is at last impossible to satisfy the demand which precisely in virtue of the concept of relation must be made, namely to win insight into the relation between the action's inner and outer conditions and into the magnitude of the ethical work which the individual man has

⁶³Perhaps the statement of DROYSEN going in the same direction (see above p. 81) is in part due to the influence of Kant's train of thought.

performed or can perform from his individual conditions. All ethical judgments are therefore imperfect and hold at most only approximately.

Kant's assumption of a norm holding for all men can also still be maintained. For the principle of relation becomes at bottom one with the principle of personality, that is, the principle that every man shall always be regarded and treated as end, never merely as means. This is satisfied when the demands upon the individual are not merely socially motivated, but also individualized according to the individual's capacity and need, so that his personality can be developed through the work he does to follow the norm.

Two of Kant's most significant ethical thoughts thus come into their full right only through application of the concept of relation.

f. Philosophy of Religion and the Concept of Relation.

The religious problem too, however one may stand towards it, exhibits a many-sided application of the concept of relation. This problem rests in all its forms on an opposition between the valuable on the one side, the valueless or value-hindering on the other, — more briefly expressed, between value and reality, in that reality exhibits both value and its opposite. The opposition here present need not — in my view — be a dualism, as has been thought⁶⁴; but it may certainly rise to a catastrophe, bring with it that tragedy of life the possibility of which no view of life can exclude, as little as one can exclude the possibility of earthquakes. And even if in the tragic, in the suffering that leads to the limit of the human, one would find the true nature of existence expressed, that would precisely confirm the significance of oppositions for the conception of life. The tragic presupposes relation in the sharpest sense, without however excluding a holding fast to the valuable. The tragic hero maintains humanity's nobility in the greatest catastrophe. But apart from the great catastrophes too, the relation between value and reality appears as the problem on which religion under all forms rests. In all forms of religion the point is what is victorious in the world, the good or the evil. Within its great fullness reality exhibits the possibility

⁶⁴LA HARPE: *La Religion comme conservation de la valeur, dans ses rapports avec la philosophie générale de Harald Høffding* (1920), p. 93 ff.

of development in various directions, and the contention between forces working in the direction of what men ascribe highest value to, and other forces working in a quite other, perhaps opposite direction, sets precisely the questions which religion in one form or another seeks to answer. It is this contention that leads the religious consciousness to form two concluding concepts, God and world, and all religious thinking is a brooding over the relation between these two concepts. In asceticism and in mysticism one seeks, now more in practice, now more in speculation, to reach beyond all relations, or (as we have seen above p. 47 Note) at least to make the one term of the relation disappear; but again and again relation forces itself forward and sets the problem anew. The need of life disturbs the ascetic, and the world of reality forces its way into the mystic's cell with its manifoldness and its unrest. One discovers at last the position of the one compass-leg, however well it was hidden.

The question is whether the valuable in the world (the valuable part of reality) is a foreign guest which has no home there, or whether it should not rather be a disclosure of existence's inmost nature. The only rational answer which can here be given (and which I have attempted to give in "Totalitet som Kategori" Kap. 5) must proceed from the close connexion between the concepts value and whole. Everything to which value is ascribed is the expression or the condition of a certain whole's subsistence and development, and the valuable arises according to the same laws that make formations of wholes (for example organisms, personalities, societies) possible. (Cf. above p. 83 f.). When now whole stands against whole, or when chaotic elements make the formation of wholes impossible, then there are present precisely those experiences which lead to a distinction between value and reality.

When the monotheistic religions regard God as the origin of all reality, the problem of evil appears with particular sharpness. Outside the origin of all reality, evil cannot have its origin. Therefore bold and profound speculation, such as Jacob Böhme's, has drawn the consequence that there must be a dark ground in God himself, out of which what conflicts with value in the world has arisen. Inner relations in God are then set in place of the outer relations at which experience stops. Still more complicated does the matter become when, as Böhme already recognized, it may be a condition for the development of the good that evil should exercise its resistance and thereby work

towards the release of forces which would otherwise lie fallow. A consideration of this kind led Goethe (*Aus meinem Leben. Fjerde Del*) to emphasize what he called the demonic in the world, and to use the motto: *nemo contra deum nisi deus ipse*.⁶⁵

Religious faith is a hope or an assurance that there must always in existence (in that which experience exhibits, or in another) be the possibility of valuable formations of wholes. It is this that is expressed by saying that religion is faith in the persistence of value. But just as wholes arise and perish, so too do values. The religious consciousness therefore comes in the course of its development to recognize that it cannot be the sensuously or empirically valuable that persists, indeed that it is at last impossible to give a concept or an image of what it is that persists. This recognition does not work its way forward without strong resistance and demands a deep resignation. It is easily veiled through the imagery religion employs.

The concept God itself contains a valuation; men's God is what they set highest. In the expression "divine" this is plainly seen; it is a determination of value. Plato already speaks of "that which makes God divine", which is that God lives constantly in the great Ideas, which men can do only now and then (*Phaedrus* p. 249 C). Plotinus goes a step further, in saying that what makes the gods divine must itself be the highest divinity (*Ennead*. V, 1, 2). The continuation of this train of thought would lead all theology back to a doctrine of value.⁶⁶ —

The expression "the persistence of value" says nothing about how much value there is in the world, or of what kind it is. On these two points lie the differences between the various religions or religious standpoints, which this is not the place to enter upon; that must be left to the history of religion and the psychology of religion. For the theory of relation it was here only the task to show that the concept of relation involuntarily asserts itself in all religious thinking, the most elementary as well as the most speculative. It is, as we saw, partly the relation between value and reality, partly the relation between eternal and perishable values, that sets the questions for religious thinking. New relations arise during the historical development of the religions and often push the underlying relations into the background. Thus the relation between

⁶⁵See further on this *Religionsfilosofi*, § 88–89.

⁶⁶Cf. with this the way in which the expression "divine" is used by a modern mystic (*Oplevelse og Tydning*, p. 89).

a religion's proper kernel and its formation in cult and dogma. In its zeal to obtain guarantees for the persistence of value, the religious consciousness has a tendency to multiply the object of faith, so that the one dogma is used to support the other. — A particularly important relation is that between religion and morality. Consciousness may be so taken up with the persistence of value and with the dogmas thought to guarantee it, that energy is drawn away from the work of discovering and bringing forth values or of maintaining the values already found. One may dwell so long at the will's boundary that one does not come to work in the domains where men need not wait for the gods. And at last, from an ethical point of view, the consideration forces itself forward that value is nonetheless not necessarily dependent on persistence. One can live in and for values even if no certainty can be won about their persistence. The divine is not dependent on time. What is valued as beautiful and good can keep its value, whatever its fate may become.

5. The Concept of Relation in Cosmology.

Does the carrying through of the concept of relation permit a rounded, even if not concluded, conception of existence? Will not every conception of a whole here show itself impossible in face of a chaos of points of view, each of which lays claim to justification? — In face of such questions there are various considerations to bring forward.

a. No orientation is possible except from a definite standpoint. But every standpoint can in turn be drawn into the discussion, because it stands in definite relations and rests on definite presuppositions. There is a constant tendency to overlook this. Sound common sense, the religious world-views, and all too often natural-scientific and philosophical systems as well, have proceeded on the assumption that there are standpoints whose grounding cannot in turn be demanded, and whose justification cannot be discussed; what is the basis for a world-view is itself not to be conditioned. Every world-view has its primal phenomenon, to use Goethe's expression, and Goethe expressly asserted that the primal phenomenon cannot be explained, but must be regarded as an absolute last. But wherever one may flee, the concept of relation will

be able to assert itself, and there may arise that unrest which is inseparable from all working thought. The idylls of thinking are always exposed to dissolution and can only be provisional resting places. In the most recent time one is beginning to become clear that a rational explanation of all parts or sides of existence from a single primal phenomenon is not possible. But one then often maintains that here at last a choice must be made between two possibilities: between a mathematical-mechanical and a psychological-qualitative primal phenomenon. Natural science approaches more and more the mathematical ideal of a world of identities or substitutions, where differences of time and quality play no part, while the human sciences (or, as some prefer to call them, the sciences of culture) maintain the reality of differences of quality and stages of development which exclude reduction to identity. Between the two types thus determined the choice is supposed to lie. In this direction Wundt, James Ward and Henri Bergson have expressed themselves. But that such a choice should be necessary would presuppose that an either—or is here present, that there thus exist no other types of existence than the two indicated. The circumstance that identity, quality and temporality cannot be reduced to one does not justify an either—or. Irreducibility is one thing, absence of all other possibilities another. In face of an assertion of an either—or there is established here the significance of Spinoza's remarkable doctrine of the infinitely many attributes under which existence appears, but of which our experience shows us only two — precisely those two types between which the choice is supposed to lie. To a far higher degree than he is conscious of, Spinoza here points to the limitation of our knowledge.⁶⁷ When Spinoza, in spite of the manifoldness of points of view whose possibility and justification he acknowledged, nonetheless maintained a doctrine of unity, it was because he too had his primal phenomenon, which he was convinced could not be denied from any of the many points of view. He held to the fact that we have a science — that rational understanding of the world is possible to the extent that it is. It was this fact that he projected outward as an absolute reality and called the substance in everything. And it is at any rate a foothold making it possible to discuss the various points of view that may assert themselves, particularly also with regard to whether their mutual relation is contradictory or

⁶⁷ Spinoza's *"Ethica"*, p. 22–24.

merely contrary. The very limits to which human knowledge is always subject can be determined only by scientific thinking. And here it shows itself, as remarked several times in the foregoing, that the necessity of relations can be both a barrier and a spur for knowledge. Where we can find no definite relations, we have no knowledge either. Our seeking therefore does not cease; we know what we lack: new positions for our compass-legs. Therefore we end with a questioning.

It has now even been thought that the carrying through of the concept of relation would make all truth impossible. Bergson distinguishes between relative and limited knowledge; the first distorts the nature of its object, the second leaves it untouched, in that it grasps only a part of it.⁶⁸ But with the relations the problems too fall away; there become neither definite questions to put nor definite answers to give. Consistently Bergson therefore points to an immediate intuition as the only way by which truth can be attained and the inmost essence of existence be given. In the closer carrying through of this, however, he makes use of constant inferences from analogy — and to that extent of relations too, since analogy means likeness of relation.⁶⁹ An example of how relation forces its way in precisely when one has thought to be able to exclude it! — The most important objection to Bergson's philosophy is from a philosophical point of view not the one Poincaré has somewhere expressed in the sentence: *Le monde Bergsonien n'a pas des lois!* — but this, that this philosophy has no problems. It is really a glorification of the problemless state.

The life of human thought expresses itself in the relations it is able to bring out and to determine. Every such determination means a victory over a chaos. In a chaotic series of difference no definite relations whatever are demonstrable; absolute interchangeability prevails. That the bringing out and determining of relations is thought's essential property follows, as mentioned several times in the foregoing, from relation being correlative to synthesis. The unity and energy of thought, of consciousness, express themselves in the relations that let the individual objects appear in their peculiarity. A comprehending together by which no relations are set is a contradiction. Even in dreams the law of determination through relation prevails, however much they may approach chaos.

⁶⁸ *Bulletin de la société française de Philosophie*. VIII, p. 341.

⁶⁹ *Henri Bergson's Filosofi*, p. 54.

Nor is it in and for itself an imperfection in our knowledge that at different times and in different places, according to the different points of departure, it comes to different results. Thereby there is merely set a new problem, a new task for knowledge, namely to find out how the different points of departure and thereby the different results became possible. In the foregoing we have seen how the physicist and the historian recognize the possibility of different standpoints and of the different conceptions following from them. It is a task which of course cannot always be solved, but which necessarily presents itself, since truth must be able to explain both itself and the errors; that belongs to complete truth. Plato already saw this when he (Republic X, p. 602 C–603 A) shows that when different observers apprehend the same thing in different ways as regards magnitude and figure, one can by counting and measuring arrive at a conception from which the various individual conceptions can be explained. Galileo was inspired by this thought in the founding of his doctrine of motion. But he was embarrassed by what his imperfect telescope showed him of the phenomenon now called Saturn's ring. The phenomenon appeared now as edges or corners on the planet, now as two different subsidiary planets. At last Galileo gave up all inquiry about it. Later Huyghens with his improved telescopes saw plainly that Saturn was surrounded by a ring, and showed how earlier conceptions could be explained from this.⁷⁰ It is the triumph of thought in every domain when a "perfect" knowledge can be won from which the "imperfect" conceptions can be explained. When Spinoza teaches (Eth. II, 36. Dem.) that our "imperfect" presentations arise with the same necessity as the "perfect" ones, it is precisely by the "perfect" (i.e. more perfect) presentation that this necessity is recognized. That is the meaning of Spinoza's famous proposition: *veritas est norma sui et falsi* (Eth. II, 43. Schol.).

Thought thus sets its boundary problems in virtue of its own law, and it is in virtue of this same law that it explains its errors.

b. But the very counting and measuring to which Plato pointed, and of which Galileo made use in principle, is after all due to human mental activity, and what right have we to ascribe validity to its results? We wish after all to understand the world as it is, and not as it presents itself to human calculation, even where the latter attempts

⁷⁰HOEFER: *Histoire de l'Astronomie*, p. 445–447.

to adopt ever so essential points of view.

Already above (p. 58 ff.) I have, in face of such a consideration, impressed that the concept of relation must be applied not merely between objects among themselves, but also between the objects and the knowing subject, or, more correctly expressed, that the knowing subject is itself an object. "The percipient event" too, to use Whitehead's expression, belongs among the events in the world. We have no right to assert in advance an absolute opposition between object and subject. Even if one seeks to transform all science into formal science and reckons everything that cannot be brought into this form among "anthropomorphisms", these anthropomorphisms are after all a fact which also demands its scientific investigation.

At the basis of those questions there really lies the popular and dogmatic definition of truth as our knowledge's agreement with reality. This static concept of truth is, as has often been shown, self-contradictory. Reality does not stand fully finished, so that we need only go and view it. It first comes into being for us through strict mental labour in observation and thinking. The history of this labour is the history of truth. It serves to characterize existence itself that for a certain stage of the development of human knowledge it had to appear in a certain definite way. We need only, for example, enter into all the presuppositions of the Middle Ages' rejection of antipodes in order to see what truth had to stand as necessary at this stage. What is reality for men at a given stage depends on how far the mental labour at this stage has advanced. Later this reality may stand as an imperfect first attempt. The eternity of truth rests on the continuity of mental labour.

It is not an intellectually aristocratic train of thought to which we are hereby led. Precisely according to the concept of relation there is truth in every honest striving after clarity. We do not ourselves create the presuppositions from which we think, and we do not even become fully conscious of them. There is here an analogy between the intellectual and the ethical life (see above p. 81 ff.). In the necessity with which the "imperfect" presentations appear, eternal truth is itself present. And often it has even been a condition for progress in the knowledge of truth that full earnest was made of the (as it later showed itself) false presuppositions. As William Hamilton has said: a living untruth is better than a dead truth.

Beyond the constant striving to which the dynamic concept of truth points we do not get. Our thought cannot leap over its own shadow. Our last foothold is the working life of thought itself in its constant discovering and determining of relations. A conclusion is here not possible.

Bacon and Comte maintained that man is to be explained by the world, not the world by man. But here there is no dilemma. We have, as we have seen (above p. 60 ff.), never an absolute subject over against an absolute object, but indeed an objectively determined subject (S_o) over against a subjectively determined object (O_s), and they both demand a searching characterization. And when we then compare them, this comparison must be undertaken by a subject, so that the problem constantly recurs. He who would explain man from the world must not forget that the world is known only through human inquiry, and that all investigation of the relation between world and man is instituted by a human thought.

c. When we believe that existence is more comprehensive than what our thought at any time or in any form can comprehend, this is grounded in our experiencing again and again how new relations and thereby new questions emerge and demand determinations. We have therefore, as shown above, no right to believe that existence is exhausted by the two primal phenomena or attributes which can briefly be denoted as mind and matter. It is the teasing thing about the concept of relation that it constantly shows us a hole in our systems. Even if we have formed the most beautiful totalities in our thought, the question emerges what this totality stands in relation to.

Investigation of the concept of totality in its significance for the world-view ends, as I have elsewhere (*Totalitet som Kategori*, Kap. 6) sought to show, in two problems: that of the possibility of saying anything about existence as a whole, and that of the possibility of determining the position of the valuable within such a whole — thus in a cosmological and in a religious problem. When the demand for grounding is made sharply and definitely, these problems can be solved only by the way of analogy. Of this a critical investigation of cosmological and religious notions also testifies. (Cf. the epistemological part of my *Religionsfilosofi*.) For the philosopher such analogies will lie nearest as are drawn from the working life of thought and will, whose history is for us the history of truth and goodness. It is due to a mystical need

in Spinoza when he projected outward the connexion of thought which for him was the primal phenomenon as a resting substance. Our last analogies must rely on our most significant experiences, and these have labour, striving and struggle for their content, corresponding to the constant contention between our two most fundamental categories, synthesis and relation. Without the appearance of oppositions and striving to overcome them, existence cannot, philosophically seen, be thought. Among these oppositions the opposition between the valuable and the value-hindering asserts itself, practically seen, most strongly.

If no grounded assurance can be won that all oppositions and inhibitions can be harmonized, the analogy with the working life of thought and will can nonetheless afford a poetic and symbolic expression for the assurance that the labour in which we are placed is not an isolated feature in reality; for that, relation and the struggle with relations hang far too closely together with the knowledge of existence which we are at all able to win.

d. Only in the resting form of the image can we, for a while, get beyond the constant unrest which by reason of the concept of relation is characteristic of every waking thought. It is therefore also to the world of images that he who seeks rest and peace takes his refuge. A famous Danish theologian has stated that without the use of images we should come out into the void, in that we should have to hold to some poor concepts that could give no light. Yes, the philosopher has only his poor concepts to hold to. But if these concepts under constantly renewed reflection show themselves to be grounded in the nature of the life of thought and of experience, then he may have the right to believe that the pulse of existence beats in the best and clearest thoughts men can form, and perhaps does not beat most faintly where new questions are set in virtue of the concept of relation. And he reserves his judgment on the analogies lying at the basis of the formation of images in the individual case. As little as any thought will any image be able completely to express the inconcludable existence of which we are members. In the fact that every parable limps, the concept of relation announces itself again and again.

In Obstfelder's "En Præsts Dagbog" the struggle between thought and image is depicted in a profound and atmospheric way, as it is often waged in a yearning mind.

The poet is clear — as the philosopher too can be — that the highest dwells “further within us than we ourselves can penetrate”. But he is not satisfied with this conviction. He wishes to have it out before him, to “come outside that which is within us”, to be able to say thou to it, to have an image of it. If he had succeeded in reaching this goal, he would have made the experience that no image suffices, but that a constant need for new images asserts itself, just as the investigator constantly discovers new relations by which his hitherto formed concepts are burst asunder.

The world of poetry stands open to all, also to the philosopher. But the philosopher cannot acknowledge any image formed once for all. And he will be clear that when he has passed over into the world of poetry, the thoughts fall silent, just as for the Anacreontic poet “the cares fall silent” when he grasps his cup. There is constantly required new labour, both with images and with thoughts, in order that life may come into its own. Perhaps what an Egyptian sage called “a yearning mind, whose words are all hidden” is the most inward and highest form of the life of the spirit.

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